

AN INTRODUCTION TO MONSTROUS MOONSHINE

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"I can explain what monstrous
moonshine is in one sentence.
It is the voice of God."

Simon P. Norton

ABSTRACT. We present an overview of theory underlying monstrous moonshine with a special focus on affine Lie algebras. Explicit examples and computations will be given throughout.

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1. INTRODUCTION

Consider the following two seemingly unrelated sequences. First, the coefficients of the j -invariant, an example of a *modular function*,

$$j(q) = q^{-1} + 744 + 196884q + 21493760q^2 + \dots$$

and second, the dimensions of irreducible representations of the *Monster group* M .

$$1, 196883, 21296876, \dots$$

These look eerily similar. In fact, we can write the first sequence in terms of simple sums of terms in the second sequence

$$196884 = 196883 + 1$$

$$21493760 = 21296876 + 196883 + 1$$

and so on for all higher powers of q . This connection, if it was more than just a coincidence, seemed so bizarre that it got the nickname *monstrous moonshine*. However, as it turns out, it is actually just the surface of an incredibly deep connection between many disparate areas of mathematics (and, as it happens, physics).

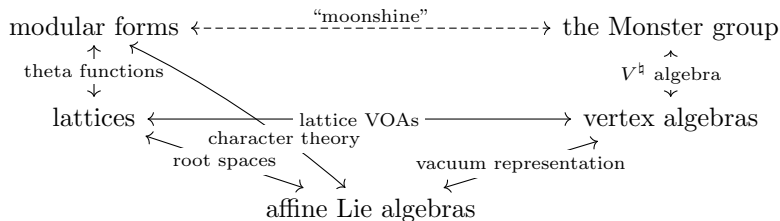


FIGURE 1. The monstrous moonshine connection.

2. THETA FUNCTIONS AND MODULAR FORMS

As with many topics we will cover, the theory of modular forms is far too deep for us to do it justice without becoming too distracted from the main goals of our paper. However we will present the basic definitions as well as certain particular facts that will be important to our later explorations. Details can be found in any standard text on modular forms, we recommend [9] and [10].

2.1. Basics. Modular forms are essentially functions on the complex upper half plane that have interesting transformation properties under the action of $SL_2(\mathbb{Z})$. The Fourier coefficients of these functions can often be interpreted as “counting” things from number theory or other areas of mathematics. If $\mathbb{H}$ is the upper half plane, we define the action of $SL_2(\mathbb{Z})$ by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}(\tau) = \frac{a\tau + b}{c\tau + d}.$$

Modular forms are essentially functions that behave in certain simple but interesting ways under this action.

Definition 2.1. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$. Given $k \in \mathbb{Z}$, the operator $[\gamma]_k$ sends $f : \mathbb{H} \rightarrow \mathbb{C}$ to the function $f[\gamma]_k$ where

$$f[\gamma]_k(\tau) = (c\tau + d)^{-k} f(\gamma(\tau)).$$

Functions that are invariant under the action of $SL_2(\mathbb{Z})$, i.e. $f = f \circ \gamma$, can also be characterized by $f = f[\gamma]_0$ for all $\gamma \in SL_2(\mathbb{Z})$. We call such functions *modular functions*.

Definition 2.2. We say that f is a *modular form of weight k* with respect to Γ , where Γ is a finite index subgroup of $SL_2(\mathbb{Z})$, if

$$f = f[\gamma]_k \quad \forall \gamma \in \Gamma$$

and $f[\alpha]_k$ is holomorphic on $\mathbb{H}$ and at infinity for all $\alpha \in SL_2(\mathbb{Z})$ (see (2.3) or Proposition 2.4 for an exact meaning of holomorphic at infinity, but essentially the function must be bounded as $\Im(\tau) \rightarrow \infty$).

We will typically only worry about modular forms with respect to the full modular group, or subgroups which contain the matrix

$$\mathcal{T} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

In general all the subgroups one cares about, the so-called *congruence subgroups*, contain a matrix of the form $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} = \mathcal{T}^n$. Regardless, assuming $\mathcal{T} \in \Gamma$ we have $f(\tau) = f[\mathcal{T}]_k(\tau) = (0\tau + 1)^k f(\tau + 1) = f(\tau + 1)$. Starting now and throughout the rest of the paper let $q = e^{2\pi i\tau}$. The fact that $f(\tau) = f(\tau + 1)$ makes

$$g(q) = f\left(\log(q)\frac{1}{2\pi i}\right)$$

a well-defined function from the punctured unit disk to $\mathbb{C}$ (because the function $e^{2\pi i\tau}$ maps $\mathbb{H}$ to the punctured disk). Furthermore, the statement that $f[\alpha]_k$ is holomorphic at ∞ means precisely that $g(q)$ has a holomorphic extension to the point 0. Thus we can write $g(q)$ as a Taylor series

$$(2.3) \quad f(\tau) = g(q) = \sum_{n=-N}^{\infty} a_n q^n = \sum_{n=-N}^{\infty} a_n e^{2\pi i n \tau}$$

which is called the Fourier development of f . We will use the following proposition.

Proposition 2.4. *Suppose $f : \mathbb{H} \rightarrow \mathbb{C}$ is holomorphic and satisfies $f[\gamma]_k = f$ for all γ in some large subgroup of $\mathrm{SL}_2(\mathbb{Z})$. If we can also write*

$$f(\tau) = \sum_{n=0}^{\infty} a_n q^n$$

and for all $n > 0$ we have $|a_n| \leq Cn^r$ for some constants C and r , then $f[\alpha]_k$ is holomorphic at ∞ for all $\alpha \in \mathrm{SL}_2(\mathbb{Z})$ (and so f is a modular form).

It is easy to show that $[\cdot]_k$ is multiplicative, i.e. if $\gamma_1, \gamma_2 \in \mathrm{SL}_2(\mathbb{Z})$ then

$$(f[\gamma_1]_k)[\gamma_2]_k = f[\gamma_1\gamma_2]_k$$

thus to show that a function f is invariant under some subgroup Γ , it suffices to show that it is invariant under the generators of that subgroup. In particular, it is well-known that $\mathrm{SL}_2(\mathbb{Z})$ is generated by the matrix $\mathcal{T}$ from above as well as

$$\mathcal{S} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Thus to show that $f(\tau)$ is a modular form for the full modular group, it suffices to investigate its behavior under the transformations $\tau \rightarrow \tau + 1$ and $\tau \rightarrow -\frac{1}{\tau}$ respectively. Now let's get some examples.

2.2. Theta Functions. The most important examples of modular forms for our purposes will be theta functions, which essentially create modular forms out of lattices. Thus we briefly introduce lattices:

Definition 2.5. A lattice Λ in $\mathbb{R}^n$ is a subset of $\mathbb{R}^n$ of the form

$$e_1\mathbb{Z} \oplus \cdots \oplus e_n\mathbb{Z}$$

where $e_1, \dots, e_n$ are linearly independent, i.e. the matrix A with columns $e_1, \dots, e_n$ is non-singular. The *discriminant* of Λ is

$$\mathrm{disc} \Lambda = \det A.$$

The *dual lattice* to Λ is

$$\Lambda^* = \{v \in \mathbb{R}^n : (v, \ell) \in \mathbb{Z} \quad \forall \ell \in \Lambda\}.$$

Finally, Λ is *even* if $(\ell, \ell) \in 2\mathbb{Z}$ for all $\ell \in \Lambda$ and *self-dual* if $\Lambda = \Lambda^*$.

We will use the following two notations for theta functions (note that many different sources use different notation depending on what aspect of theta functions they care about).

Definition 2.6. Let $m, n \in \mathbb{Z}$. Then let $\Theta_{m,n}(z, \tau, t)$ be a function of three variables with $z, t \in \mathbb{C}$ and $\tau \in \mathbb{H}$ with

$$\Theta_{n,m}(z, \tau, t) = e^{2\pi i m t} \sum_{\ell \in \frac{n}{2m} + \mathbb{Z}} e^{2\pi i m \ell z} q^{m \ell^2}$$

Now let Λ be an even self-dual d -dimensional lattice. Then let

$$\Theta_{\Lambda}(\tau) = \sum_{\ell \in \Lambda} q^{(\ell, \ell)/2}.$$

Note that $\Theta_{m,n}(0, \tau, 0)$ is also essentially a lattice theta function over a one-dimensional lattice, with some small shift to the lattice and a scaling by $2m$ in the exponent. Indeed, if $n = 0$ these definitions coincide, and in particular

$$\Theta_{0,1}(0, \tau, 0) = \Theta_{\sqrt{2}\mathbb{Z}}(\tau)$$

which we will sometimes just refer to as $\Theta(\tau)$.¹

To show the modular properties of these functions we will use the following formula:

Theorem 2.7 (Poisson Resummation). *Let $f : \mathbb{R}^n \rightarrow \mathbb{C}$ be a Schwartz function, i.e. $f(x)$ and all of its partial derivatives decay as $o(\|x\|^{2k})$. Then for any lattice Λ in $\mathbb{R}^n$ we have*

$$\sum_{\ell \in \Lambda} f(\ell) = (\text{disc } \Lambda)^{-1/2} \sum_{\ell \in \Lambda^*} \hat{f}(\ell)$$

where $\hat{f}$ is the standard Fourier transform

$$\hat{f}(y) = \int_{\mathbb{R}^n} f(x) e^{2\pi i (x, y)} dx.$$

Proposition 2.8. *The functions $\Theta_{m,n}(0, \tau, 0)$ and $\Theta_{\Lambda}(\tau)$ are modular forms (up to a phase).*

Proof. We will work out the lattice theta function in detail (assuming facts such as the convergence of these sums). Since Λ is even, all the exponents on $q = e^{2\pi i \tau}$ are integers and so invariance under $\mathcal{T}$ (i.e. the transformation $\tau \rightarrow \tau + 1$) is immediate. For the transformation under $\mathcal{S}$, i.e. $\tau \rightarrow -\frac{1}{\tau}$, we have

$$\Theta_{\Lambda}\left(-\frac{1}{\tau}\right) = \sum_{\ell \in \Lambda} e^{-\pi i (\ell, \ell)/\tau} = (\text{disc } \Lambda)^{-1/2} \sum_{\ell \in \Lambda^*} \hat{f}(\ell)$$

and by assumption Λ is self-dual. It is easy to show that this implies $\text{disc } \Lambda = 1$, so we just have $\sum_{\ell \in \Lambda} \hat{f}(\ell)$. It suffices to compute the Fourier transform of $f(\ell) = e^{-\pi i (\ell, \ell)/\tau}$. Choose some orthonormal basis so that $x \in \mathbb{R}^n$ has coordinates

¹Technically $\sqrt{2}\mathbb{Z}$ is not a self-dual lattice so it is an abuse of notation to write $\Theta_{\sqrt{2}\mathbb{Z}}$

$(x_1, \dots, x_n)$. We have

$$\begin{aligned}\hat{f}(y) &= \int_{\mathbb{R}^n} e^{-\pi i(x,x)/\tau} e^{2\pi i(x,y)} dx \\ &= \int_{\mathbb{R}^n} e^{-\pi i(x_1^2 + \dots + x_n^2)/\tau} e^{2\pi i(x_1 y_1 + \dots + x_n y_n)} dx \\ &= \prod_{i=1}^n \int_{-\infty}^{\infty} e^{-\pi i x_i^2/\tau} e^{2\pi i x_i y_i} dx_i\end{aligned}$$

We can then complete the square in the exponent and this becomes essentially a standard Gaussian integral in 1 dimension. We find

$$\int_{-\infty}^{\infty} e^{-\pi i x_i^2/\tau} e^{2\pi i x_i y_i} dx_i = \left(\frac{\tau}{i}\right)^{\frac{1}{2}} e^{-\pi \tau y_i^2/i} = (-i\tau)^{\frac{1}{2}} e^{\pi i \tau y_i^2}$$

and so

$$\Theta_{\Lambda}\left(-\frac{1}{\tau}\right) = \sum_{\ell \in \Lambda} f(\ell) = (-i\tau)^{\frac{n}{2}} \sum_{\ell \in \Lambda} e^{\pi i \tau(\ell, \ell)} = (-i\tau)^{\frac{n}{2}} \Theta_{\Lambda}(\tau)$$

so Θ_{Λ} has the desired transformation properties with weight $\frac{n}{2}$ up to the phase $(-i)^{n/2}$. Note that if n is divisible by 8, then this phase is just 1. If we want to write

$$\Theta_{\Lambda}(\tau) = \sum_{k=0}^{\infty} a_k q^k$$

then by grouping terms with the same exponent we find that a_k is the number of $\ell \in \Lambda$ such that $(\ell, \ell) = 2k$. This is clearly less than the number of lattice points contained within a radius of $\sqrt{2k}$ from the origin, which grows as k^n . Thus indeed $|a_k| \leq Ck^n$ for some constant C , so by [Proposition 2.4](#), Θ_{Λ} is a modular form of weight $\frac{n}{2}$.²

Note that if Λ is not even, our argument for the $\mathcal{S}$ transformation remains valid, however now the exponents are only necessarily half-integral, so instead of invariance under $\mathcal{T}$ we get invariance under $\mathcal{T}^2$ and the subgroup $\langle \mathcal{T}^2, \mathcal{S} \rangle$.

The picture for $\Theta_{n,m}$ is slightly different, although the argument goes pretty much the same (also using Poisson resummation and completing the square to evaluate the Gaussian). We will just expand $\Theta_{0,1} = \Theta$ in detail. Again invariance under $\mathcal{T}$ is immediate. However, the lattice $\sqrt{2}\mathbb{Z}$ is not self-dual, so our second argument does not work. Instead consider the transformation $\tau \mapsto -\frac{1}{4\tau}$. We get

$$\Theta\left(-\frac{1}{4\tau}\right) = \sum_{k \in \mathbb{Z}} e^{-\pi k^2(i/2\tau)} = \sum_{k \in \mathbb{Z}} \sqrt{\frac{2\tau}{i}} e^{2\pi i \tau k^2} = \sqrt{-2i\tau} \Theta(\tau).$$

²Due to the ambiguities of the square root in $\mathbb{C}$ we initially only defined modular forms for integer weights, but one can either take the principle branch of the square root or just take this to mean that Θ_{Λ}^2 is a modular form of weight n .

The transformation $\tau \mapsto -\frac{1}{4\tau}$ corresponds to the matrix $\begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix}$ which is notably not in $\mathrm{SL}_2(\mathbb{Z})$. However, the matrix $\begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$ is, and we have

$$\begin{aligned} \Theta\left(\frac{1}{4\tau+1}\right) &= \Theta\left(-\frac{1}{4(-1/(4\tau)-1)}\right) = \sqrt{2i\left(\frac{1}{4\tau}+1\right)}\Theta\left(-\frac{1}{4\tau}-1\right) \\ &= \sqrt{2i\left(\frac{1}{4\tau}+1\right)}\Theta\left(-\frac{1}{4\tau}\right) = \sqrt{2i\left(\frac{1}{4\tau}+1\right)}(-2i\tau)\Theta(\tau) \\ &= \sqrt{4\tau+1}\Theta(\tau). \end{aligned}$$

The same argument as above ensures it is holomorphic at ∞ , so $\Theta(\tau)$ is a modular form of weight $\frac{1}{2}$ with respect to the subgroup

$$\left\langle \mathcal{T}, \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix} \right\rangle.$$

This subgroup is sometimes referred to as $\Gamma_0(4)$. □

Proposition 2.9. *We will also use the following identity:*

$$\Theta_{n,m}(\tau, z, u) \Theta_{n',m'}(\tau, z, u) = \sum_j d_j^{(m,m',n,n')}(q) \Theta_{n+n'+2mj, m+m'}(\tau, z, u)$$

where the sum is over $j \in \mathbb{Z} \bmod (m+m')\mathbb{Z}$, and

$$d_j^{(m,m',n,n')}(q) = \sum_k q^{mm'(m+m')k^2/2} \quad \left(k \in \mathbb{Z} + \frac{m'n - mn' + 2jmm'}{2mm'(m+m')}\right).$$

The proof is a long but straightforward computation, so we refer to the reader to [11].

2.3. The j Modular Invariant. A very natural object to study in this discussion is the quotient space $Y(1) = \mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ (or in general $\Gamma \backslash \mathbb{H}$ for Γ a subgroup of $\mathrm{SL}_2(\mathbb{Z})$). Indeed, a function which is invariant under the action of $\mathrm{SL}_2(\mathbb{Z})$ can equivalently be defined as a function on this quotient space. To understand the topology of this quotient, it is helpful to find the *fundamental domain* of this action, i.e. some nice subset of $\mathbb{H}$ that contains exactly one point in each $\mathrm{SL}_2(\mathbb{Z})$ equivalence class (up to some boundary identification).

Proposition 2.10. *The fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$ is*

$$\mathcal{D} = \left\{ \tau \in \mathbb{H} : |\Re(\tau)| \leq \frac{1}{2} \text{ and } |\tau| \geq 1 \right\}$$

where points on the boundary (i.e. points τ such that $|\tau| = 1$ or $|\Re(\tau)| = \frac{1}{2}$) are identified by reflection across the imaginary axis.

We will not give the proof here, but it is not hard to intuit that any point in $\mathbb{H}$ can be uniquely mapped into this set. Recall the generators $\mathcal{S}, \mathcal{T}$ for $\mathrm{SL}_2(\mathbb{Z})$. Given a point τ we can essentially use $\mathcal{S}$ to move it outside the unit circle, and then $\mathcal{T}$ to shift it so its real part lies between $-\frac{1}{2}$ and $\frac{1}{2}$, possibly having to repeat this process multiple times.

Thus $Y(1)$ can be pictured as this domain, with the vertical line segments $\Re(\tau) = -\frac{1}{2}$ and $\Re(\tau) = \frac{1}{2}$ glued together, and opposite sides of the arc $|\tau| = 1$ glued together, which looks like some sort of strange cone. The next natural step is to compactify this cone by “sealing off” the top. To do this, we want to add a point

at infinity. We then also need to add the possible $\mathrm{SL}_2(\mathbb{Z})$ images of ∞ , which are precisely $\mathbb{Q}$.

Definition 2.11. Let $\mathbb{H}^* = \mathbb{H} \cup \mathbb{Q} \cup \{\infty\}$. Then the modular curve $X(1)$ is

$$X(1) = \mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}^*.$$

By imbuing $\mathbb{H}^*$ with an appropriate topology and $X(1)$ with the quotient topology, we find

Proposition 2.12. *$X(1)$ is a compact Hausdorff Riemann surface.*

Holomorphic functions on $X(1)$ are precisely modular forms of weight 0. However, it turns out all such functions are simply constant. Thus, rather than study weight 0 modular forms with respect to $\mathrm{SL}_2(\mathbb{Z})$, we broaden our scope to meromorphic functions on $X(1)$, i.e. $\mathrm{SL}_2(\mathbb{Z})$ invariant meromorphic functions on $\mathbb{H}$ with a constraint on their behavior around ∞ (the condition is precisely that the function g from (2.3) is meromorphic at 0).

Returning the topology of $X(1)$, now that it has been “sealed” at the top we hope it is homeomorphic to a sphere, with distinguished points at $e^{2\pi i/3}$, i , and ∞ . The natural map from $X(1)$ to the Riemann sphere $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ is provided by the j function.

Definition 2.13. The *modular invariant* j is the unique (up to scaling) meromorphic function $j : X(1) \rightarrow \hat{\mathbb{C}}$ such that $j(e^{2\pi i/3}) = 0$ and $j(\infty) = \infty$, both with degree 1, and it has no other poles or zeroes.

Since the pole at infinity has degree 1, when considered as a function from $\mathbb{H}$ to $\mathbb{C}$, j takes the form

$$j(\tau) = a_{-1} \frac{1}{q} + \sum_{n=0}^{\infty} a_n q^n$$

so we choose the scaling of j such that $a_{-1} = 1$. This uniquely defines j . It turns out the first few terms in this expansion are

$$j(\tau) = \frac{1}{q} + 744 + 196884q + 21493760q^2 + \cdots$$

Proposition 2.14. *j is a homeomorphism between $X(1)$ and $\hat{\mathbb{C}}$.*

Proof. Being meromorphic with codomain $\hat{\mathbb{C}}$, j is holomorphic as a map between the Riemann surfaces $X(1)$ and $\hat{\mathbb{C}}$. Thus j is a non-constant holomorphic map between Riemann surfaces, so by the open mapping theorem of complex analysis, $j(X(1))$ must be open. However, $X(1)$ is compact so $j(X(1))$ must be compact as well (and so closed) in $\hat{\mathbb{C}}$. The only nonempty open and closed subset of $\hat{\mathbb{C}}$ is the Riemann sphere itself, so j is surjective.

We take from the theory of compact Riemann surfaces that every holomorphic map $f : X \rightarrow Y$, where X and Y are compact Riemann surfaces, has a well-defined degree d , i.e. if at each point $x_0 \in X$, the function $f(x) - f(x_0)$ has a zero of degree e_{x_0} , then the sum

$$\sum_{x \in f^{-1}(y)} e_x$$

is some constant value d for all $y \in Y$. In particular, we are given that j has a single 0 of degree 1, so by investigating $j^{-1}(0)$ we conclude $d = 1$, which means there must be precisely one point in $j^{-1}(y)$ for all $y \in \hat{\mathbb{C}}$, i.e. f is injective.

Since $X(1)$ is compact and $\hat{\mathbb{C}}$ is Hausdorff, j is indeed a homeomorphism. $\square$

Since we defined $j(\infty) = \infty$, j restricts to a function from $Y(1)$ to $\mathbb{C}$, which is also a homeomorphism (and then to a holomorphic function from $\mathbb{H}$ to $\mathbb{C}$). Hopefully the importance of the j function is already apparent, however the following result shows just how central it is to the study of modular forms and modular curves:

Theorem 2.15. *The field of meromorphic functions on $X(1)$, i.e. weight 0 meromorphic modular forms, is precisely $\mathbb{C}(j)$, i.e. rational functions of j .*

Proof. Let f be a meromorphic function on $X(1)$. Considered as a function from $\mathbb{H}$ to $\mathbb{C}$, suppose it has zeroes $z_1, \dots, z_m$ and $p_1, \dots, p_n$ listed with multiplicity (our above fact about the degree of f tells us that $f^{-1}(0)$ and $f^{-1}(\infty)$ must be finite sets). Consider the function

$$g : \mathbb{H} \rightarrow \mathbb{C}, \quad g(\tau) = \frac{\prod_{i=1}^m (j(\tau) - j(z_i))}{\prod_{i=1}^n (j(\tau) - j(p_i))}.$$

so that $g(\tau)$ and $f(\tau)$ have the same zeroes and poles with the same multiplicities (note the importance of the fact that the zero of j is simple). It is a further fact that for a function between compact Riemann surfaces, the number of zeroes minus the number of poles must be 0 (counted with multiplicity), so f and g must also have the same behavior at ∞ . Therefore $f/g : X(1) \rightarrow \hat{\mathbb{C}}$ has no zeroes or poles. By our earlier argument, the only non-surjective maps between compact Riemann surfaces are constant, so f/g is constant and the proof is complete. $\square$

3. LIE ALGEBRAS AND ROOT SPACES

Before we introduce the concept of an affine Lie algebra, we must first review the theory of finite-dimensional Lie algebras.

3.1. Semisimple Lie Algebras.

Definition 3.1. A Lie algebra (over $\mathbb{R}$ or $\mathbb{C}$) is a vector space $\mathfrak{g}$ equipped with a bilinear map called the Lie bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying

$$[x, x] = 0 \quad (\text{antisymmetry})$$

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0 \quad (\text{Jacobi identity})$$

Lie algebras over $\mathbb{R}$ arise in physics when discussing “infinitesimal actions”. (Mathematically, this is described as being in the tangent space of the identity of a *Lie group*.)

Example 3.2. An “infinitesimal 3-dimensional rotation” is an element of the Lie algebra $\mathfrak{so}(3) \simeq \mathfrak{su}(2)$ of traceless anti-Hermitian 2-by-2 matrices. These generate elements of $SO(3)$, three-dimensional rotations.

Example 3.3. While not finite-dimensional, the Virasoro algebra (Section 6.3) is related to the Lie algebra of infinitesimal conformal transformations in two dimensions.

We can then define

Definition 3.4. An *ideal* of a Lie algebra is a subalgebra $\mathfrak{h}$ satisfying $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$

Definition 3.5. A *simple Lie algebra* is a non-abelian Lie algebra $\mathfrak{g}$ (that is, the Lie bracket is not the zero map) with no ideals other than 0 and $\mathfrak{g}$.

Definition 3.6. A *semisimple Lie algebra* is a direct sum of simple Lie algebras.

Example 3.7. $\mathfrak{su}(2)$ is a simple Lie algebra. $\mathfrak{u}(1) \simeq \mathbb{R}$ is a Lie algebra that is not simple because it is abelian.

3.2. Root Spaces. In an introductory quantum mechanics class, students are typically introduced to *raising and lowering operators* for angular momentum.

$$[\hat{J}_z, \hat{J}_{\pm}] = \pm \hat{J}_{\pm}$$

If we notice that the generators $\hat{J}_i$ of angular momentum form the Lie algebra $\mathfrak{su}(2)$, these operators can be derived simply by considering the definition of a *root*.

Warning 3.8. Physicists like to use *Hermitian* generators for their Lie algebras since Hermitian operators correspond to physical observables. Mathematicians prefer *anti-Hermitian* generators as they make the formulas nicer.

Note that every element x of the Lie algebra generates an *adjoint action*

$$\begin{aligned} \text{ad}_x : \mathfrak{g} &\rightarrow \mathfrak{g} \\ y &\mapsto [x, y] \end{aligned}$$

Definition 3.9. For a semisimple Lie algebra over $\mathbb{C}$, the *Cartan subalgebra* H is a maximal abelian subalgebra such that for all $h \in H$, ad_h is diagonalizable.

Example 3.10. We can view $\mathfrak{su}(2)$ as a complex vector space, giving us the *complexified* Lie algebra $\mathfrak{su}(2)_{\mathbb{C}} \simeq \mathfrak{sl}(2, \mathbb{C})$ of traceless 2×2 complex matrices. Since we will mainly be working complex Lie algebras, we will occasionally continue to simply write $\mathfrak{su}(2)$ or $\mathfrak{sl}(2)$. The Cartan subalgebra is the 1-dimensional vector space of matrices of the form

$$\begin{bmatrix} z & 0 \\ 0 & -z \end{bmatrix}$$

Choose a basis $\{H_i\}$ of H . Since H is abelian, by the Jacobi identity we have for all x that

$$[H_i, [H_j, x]] = [H_j, [H_i, x]]$$

so all of the adjoint actions commute. This means that the ad_{H_i} all have a common basis of eigenvectors. Since the Cartan subalgebra is abelian, we know that all of the H_i are eigenvectors of each other's actions with eigenvalue 0. Since it is maximal, the converse also holds. Denote all other eigenvectors by $\{E^\alpha\}$, where α is a vector such that

$$[H_i, E^\alpha] = \alpha_i E^\alpha$$

α is called a *root* of the Lie algebra.

Definition 3.11. The *Cartan-Weyl basis* of a Lie algebra is the set of eigenvectors $\{H_i\} \cup \{E^\alpha\}$.

Example 3.12. For $\mathfrak{su}(2)$, choose

$$H_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The rest of the Cartan-Weyl basis is

$$E^+ = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad E^- = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$[H_1, E^\pm] = \pm 2E^\pm$$

This is where raising and lowering operators come from. They are simply the observation that $\mathfrak{su}(2)$ has roots ± 2 .

It will be show after we define a representation that if α is a root then $-\alpha$ is also a root. In each pair, we can select a *positive root* depending on H_1 .

Definition 3.13. A *simple root* is a positive root that is not the sum of other positive roots.

Simple roots necessarily form a basis for the *root space*, the span of the roots.

Example 3.14. $\mathfrak{su}(2)_{\mathbb{C}}$ will be slightly too trivial to illustrate the following discussion, so we will introduce $\mathfrak{su}(3)_{\mathbb{C}}$. It has a Cartan subalgebra with basis

$$H_1 = \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad H_2 = \frac{1}{2\sqrt{3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

and roots

$$\pm\alpha = \pm(1, 0) \quad \pm\beta = \pm\frac{1}{2}(-1, \sqrt{3}) \quad \pm\gamma = \pm\frac{1}{2}(1, \sqrt{3})$$

We can select α , β and γ as our positive roots. Note that $\gamma = \alpha + \beta$, so our simple roots are α and β .

Note that the roots are not invariant under the choice of generators H_i . In particular, we can scale H_i by some constant and the corresponding root scales accordingly. It is a convention to have the squared norm of the roots to be 2, but we can also describe the roots in an invariant way using the *Killing form*.

Definition 3.15. The Killing form is the bilinear map

$$\mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$$

$$(x, y) \mapsto \text{tr}(\text{ad}_x \circ \text{ad}_y)$$

It is invariant under automorphisms of $\mathfrak{g}$

Definition 3.16. Given simple roots a^i , the *Cartan matrix* is the matrix whose entries are

$$A^{ij} = 2 \frac{(a^i, a^j)}{(a^i, a^i)}$$

Example 3.17. Computing the Killing form reveals that the Cartan matrix for $\mathfrak{su}(3)$ is

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Theorem 3.18. *For a semisimple Lie algebra, the Cartan matrix is non-degenerate ($\det(A) \neq 0$).*

This is foreshadowing.

Proof. This follows from the Killing form being nondegenerate. That is, it defines a Euclidean inner product on the root space. Since the roots are a basis for the root space, the matrix of their inner products is nondegenerate. $\square$

Instead of looking at the ugly Cartan matrix, we can summarize the root system instead with the much prettier *Dynkin diagram*.



FIGURE 2. Dynkin diagrams for $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$, also known as A_1 and A_2 .

The vertices represent roots, and the single edges represent that the roots have inner product -1 .

3.3. Representations.

Definition 3.19. A *Lie algebra representation* is a linear map $\rho : \mathfrak{g} \rightarrow \text{End}(V)$ for a vector space V such that

$$\rho([x, y]) = \rho(x)\rho(y) - \rho(y)\rho(x)$$

We say that V is a $\mathfrak{g}$ -module.

Finite-dimensional representations of semisimple Lie algebras take a very special form. Recall that we necessarily have that the action of the Cartan subalgebra is diagonal, so there exists a basis $\{v_\lambda\}$ such that

$$H^i v_\lambda = \lambda^i v_\lambda$$

The vectors λ are called *weights*. If V is finite-dimensional the weight must satisfy an additional property

$$(3.20) \quad 2 \frac{(\lambda, a^i)}{(a^i, a^i)} \in \mathbb{Z}$$

for every root a^i . Weights that satisfy this are called *integral*. Integral weights form a lattice. We can put a partial order on this lattice.

Definition 3.21. For two weights λ, μ , λ is *higher* than μ if $\lambda - \mu$ is a linear combination of positive roots with nonnegative real coefficients.

We can also denote an integral weight as *dominant* if $(\lambda, a^i) \geq 0$ for each positive root a^i . As it turns out, this is enough to completely classify finite-dimensional representations of semisimple Lie algebras.

Theorem 3.22. (*Theorem of the Highest Weight*) [5]

- The irreducible finite-dimensional $\mathfrak{g}$ -representations are exactly the highest weight representations with dominant integral highest weight.
- For every weight there exists a representation where it is a dominant integral highest weight.

- *Two irreducible representations are isomorphic if and only if they have the same highest weight.*

Example 3.23. The integral weights for $\mathfrak{su}(2)$ are at the integers. The dominant integral weights are the nonnegative integers. Each of these corresponds to a distinct $\mathfrak{su}(2)$ -representation. The state corresponding to the weight n is known as the $spin-\frac{n}{2}$ representation.

Applying this to the adjoint representation we get

Corollary 3.24. *There exists a highest root θ .*

3.4. Characters. Recall that we can define the *character* of a group representation ρ as

$$\chi(g) = \text{tr}(\rho(g))$$

Elements of a Lie algebra are just infinitesimal elements of the corresponding Lie group, and we can go the other way with the *exponential map* $\exp: \mathfrak{g} \rightarrow G$. This lets us define the character of a highest weight representation.

$$\chi_\Lambda(h) = \text{tr}(\exp(\rho_\Lambda(h)))$$

where $h \in H$ is in the Cartan subalgebra of $\mathfrak{g}$. There are many equivalent ways of writing this. For example, since we have that H is diagonal, we also know we can write χ_Λ as a sum over eigenspaces

$$\chi_\Lambda(h) = \sum_{\lambda} \text{mult}_\Lambda(\lambda) e^{(\lambda, h)}$$

A more striking way is *Weyl character formula*. Define the *Weyl vector* as half the sum of the positive roots

$$(3.25) \quad \rho = \frac{1}{2} \sum_{\alpha^+} \alpha^+$$

Define a *Weyl reflection* to be a reflection along the line spanned by a root. These generate a finite group called the *Weyl group* W , where the *sign* of an element is the parity of the number of reflections needed to generate it. We then have

Theorem 3.26. (*Weyl character formula*)

$$\chi_\Lambda(h) = \frac{\sum_{\sigma \in W} \text{sign}(\sigma) e^{(\sigma(\Lambda + \rho), h)}}{\sum_{\sigma \in W} \text{sign}(\sigma) e^{(\sigma(\rho), h)}}$$

Note that, as written, the character is a function on the weight space, and indeed this is generally how we will think of it. However, the Killing form provides a natural identification between the Cartan subalgebra and the weight space, so this can also be thought of as a function from the Cartan subalgebra to $\mathbb{C}$, which is more akin to the character of a group representation.

Example 3.27. Let us return to $\mathfrak{su}(2)$. Recall that there is only one positive root, $+2$, so the constraint (3.20) simply becomes $\lambda \in \mathbb{Z}$ for all weights in the representation, which happens if and only if the highest weight Λ is a (non-negative) integer. Thus let R_Λ be the representation with highest weight $\Lambda \in \mathbb{Z}_{\geq 0}$. Then $-\Lambda$ must also be a weight, and we can descend from Λ to $-\Lambda$ by subtracting the only positive root. Thus the full set of weights is

$$\Lambda, \Lambda - 2, \dots, -(\Lambda - 2), -\Lambda$$

each with multiplicity 1 (because we only have 1 root to descend to any of them).

Now we will consider the characters. The Cartan subalgebra of $\mathfrak{sl}(2)$ is one dimensional spanned by H , so let $hH \in \mathfrak{h}$ for some $h \in \mathbb{C}$, so χ_Λ can be considered as a function of h . The inner product is now just multiplication, so we have

$$\chi_\Lambda(h) = \sum_{\lambda} \text{mult}_\Lambda(\lambda) e^{(\lambda, h)} = \sum_{n=0}^{\Lambda} e^{(\Lambda-2n)h}.$$

Notice that writing $z = e^h$ this becomes

$$\chi_\Lambda(h) = \sum_{\lambda} \text{mult}_\Lambda(\lambda) e^{(\lambda, h)} = \sum_{n=0}^{\Lambda} z^{\Lambda-2n}$$

which are precisely the characters of the irreducible representations of the Lie *group* $SU(2)$. Thus our definition of character is on the right track. Now we will show that the Weyl character formula also works. Again there is only one positive root, so it is easy to compute that $\rho = 1$. Since the root space is 1 dimensional, there is only a single reflection to consider, so $W = \{\pm 1\}$. Thus the Weyl character formula becomes

$$\chi_\Lambda(h) = \frac{e^{(\Lambda+1)h} - e^{(-\Lambda+1)h}}{e^h - e^{-h}} = \frac{z^{\Lambda+1} - z^{\Lambda-1}}{z - z^{-1}}$$

which is indeed equivalent to our earlier expression by the standard algebraic formula.

It is interesting to note that the Weyl character formula can be deduced using techniques from quantum field theory [1]

4. AFFINE LIE ALGEBRAS

Our goal is now to understand a slightly more general structure: affine Lie algebras. We will begin with the specific construction via central extensions of loop algebras, and then show how it is only a slight tweak of the characterization of simple Lie algebras by their root system and Cartan matrix. Once we have defined these structures, we will turn to the study of their representations and the characters of those representations, where we will see our first moonshine phenomenon.

4.1. Construction of Affine Lie Algebras. Let $\bar{\mathfrak{g}}$ be a simple Lie algebra (the reasoning behind the bar notation will become clear). Let $\bar{\Phi}$ be its roots and Cartan-Weyl basis

$$\mathcal{B} = \{H^i \mid i = 1, \dots, r\} \cup \{E^{\bar{\alpha}} \mid \bar{\alpha} \in \bar{\Phi}\}.$$

Let f_c^{ab} be its structure constants, i.e. if $\{T^c : c \in I\}$ make up the basis with some labeling then

$$[T^a, T^b] = \sum_{c \in I} f_c^{ab} T^c.$$

The loop algebra associated to $\bar{\mathfrak{g}}$ is

$$\bar{\mathfrak{g}} \otimes_{\mathbb{C}} \mathbb{C}[z^{-1}, z]$$

(where the square brackets denote polynomials in z^{-1} and z) and Lie bracket

$$(4.1) \quad [A \otimes f(z), B \otimes g(z)] = [A, B]_{\bar{\mathfrak{g}}} \otimes f(z)g(z).$$

In particular, this space has basis of the form $T^a \otimes z^n$ where $T^a \in \mathcal{B}$, which we can write as T_n^a . Then the bracket relation becomes

$$[T_n^a, T_m^b] = f_c^{ab} T_{n+m}^c.$$

(with the sum over c implicit). It is easy to check that these brackets satisfy the Jacobi relations. Thus this is indeed a Lie algebra, which we denote $\bar{\mathfrak{g}}_{\text{loop}}$. To make this an affine Lie algebra, we need to add two more operators denoted K and D . We first add K to the center of $\bar{\mathfrak{g}}_{\text{loop}}$, making it a *central extension*. In other words, for all T_n^a in the basis of $\bar{\mathfrak{g}}_{\text{loop}}$ we have

$$[T_n^a, K] = 0.$$

If K also does not show up in the right hand side of the brackets in (4.1), this would be a rather trivial extension. It turns out that there is (up to scaling) a unique way to add K into these brackets that preserves the Jacobi relations, namely

$$[T_n^a, T_m^b] = f_c^{ab} T_{n+m}^c + m\bar{\kappa}^{ab}\delta_{m,-n}K$$

where $\bar{\kappa}^{ab}$ is the entry in the Killing form matrix for the simple Lie algebra $\bar{\mathfrak{g}}$. Under the correct normalization this becomes simply

$$[T_n^a, T_m^b] = f_c^{ab} T_{n+m}^c + m\delta_{a,b}\delta_{m,-n}K.$$

Finally, we want an element in the Lie algebra that lets us distinguish between different grades, so we add an element D with brackets

$$[D, T_n^a] = nT_n^a, \quad [D, K] = 0.$$

The letter D is chosen because D acts as a derivation on $\bar{\mathfrak{g}}_{\text{loop}}$. Thus some authors do not actually include it as an element of the Lie algebra, but we find it most convenient and precise to do so.

4.2. The Root System. In order to define/identify the root system of an affine Lie algebra, we proceed analogously to the case of simple Lie algebras. There, the first step was to identify the maximal abelian subalgebra. In the affine case, this subalgebra is not unique (even up to conjugation), but if we keep in mind what we want out of this subalgebra, there is a natural choice. In particular, we want the generators which are not in the subalgebra to be eigenvectors of the generators in the algebra in the adjoint representation. Consider the Lie bracket

$$[T_n^a, T_m^b] = f_c^{ab} T_{n+m}^c + m\delta_{a,b}\delta_{m,-n}K.$$

If we want the right hand side to be a multiple of T_m^b , it is clear we need $m = 0$, so we first consider the $\bar{\mathfrak{g}}_{\text{loop}}$ -subalgebra of elements of mode 0 (which we denote $\bar{\mathfrak{g}}_{[0]}$). It is immediate from the commutation relations that this is indeed a subalgebra, and furthermore it is isomorphic to $\bar{\mathfrak{g}}$. Thus we start by choosing the maximal abelian subalgebra of $\bar{\mathfrak{g}}_{[0]}$, which has basis

$$\{H_0^i \mid i = 1, \dots, r\}.$$

Since K was by definition central, we can also add the span of K into our abelian subalgebra. Now consider the bracket

$$[D, T_n^a] = nT_n^a.$$

All elements of the basis are naturally eigenvectors for D in the adjoint representation, so we want D to be in our abelian subalgebra. Furthermore the above relation shows that D commutes with T_n^a if and only if $n = 0$. Thus we can add D into the

subalgebra we have been building, but once we have done so it is maximal, since no remaining elements of our basis commute with D .

Notation 4.2. We denote the maximal abelian subalgebra (and thus Cartan subalgebra) generated by $\{H_0^1, \dots, H_0^r, K, D\}$ by $\mathfrak{g}_0$.

From here the definition of the roots is exactly analogous to the simple case:

Definition 4.3. Given a generator T of $\mathfrak{g}$ not contained in $\mathfrak{g}_0$, the corresponding root is the tuple $(\cdot, \cdot, \cdot)$ where the first entry is an r -dimensional vector made up of the eigenvalues of T for H_0^i , the second entry is the eigenvalue for K , and the third entry the eigenvalue for D (where as usual eigenvalue refers to the eigenvalue of T as a vector in the adjoint representation). The set of roots is denoted Φ .

The generators which are not contained in $\mathfrak{g}_0$ are of two types: $E_n^{\bar{\alpha}}$ for any $\bar{\alpha} \in \bar{\Phi}$ and any $n \in \mathbb{Z}$, and H_n^i for $i = 1, \dots, r$ and $n \neq 0$. The brackets for these generators are

$$\begin{aligned} [H_0^i, E_n^{\bar{\alpha}}] &= \bar{\alpha}_i E_n^{\bar{\alpha}} & [H_0^i, H_n^j] &= 0 \\ [K, E_n^{\bar{\alpha}}] &= 0 & [K, H_n^j] &= 0 \\ [D, E_n^{\bar{\alpha}}] &= n E_n^{\bar{\alpha}} & [K, H_n^j] &= n H_n^j \end{aligned}$$

so we find that our roots take two forms, either

$$(\bar{\alpha}, 0, n) \quad \bar{\alpha} \in \bar{\Phi}, n \in \mathbb{Z} \quad \text{or} \quad (0, 0, n) \quad n \in \mathbb{Z} \setminus \{0\}.$$

The eigenvalue of D , the third entry in this tuple, is called the *grade* of the root. We again choose which roots to label positive, and come to a definition of simple roots.

Definition 4.4. The set of positive roots is

$$\Phi_+ = \{(\bar{\alpha}, 0, n) \mid n > 0 \text{ or } (\bar{\alpha} \in \bar{\Phi}_+ \text{ and } n = 0)\}$$

The *simple roots* are again positive roots which cannot be written as a positive integer combination of other positive roots.

Observation 4.5. Given a root $\bar{\alpha}$ of $\bar{\mathfrak{g}}$, we identify $(\bar{\alpha}, 0, 0) = \bar{\alpha}$. Furthermore, let $\delta = (0, 0, 1)$. Finally let $\bar{\alpha}^{(i)}$ for $i = 1, \dots, r$ be the simple roots and $\bar{\theta}$ be the highest root of $\bar{\mathfrak{g}}$. Then the simple roots of $\mathfrak{g}$ are

$$\alpha^{(0)} = \delta - \bar{\theta}, \quad \alpha^{(i)} = \bar{\alpha}^{(i)} \text{ for } i = 1, \dots, r.$$

The fact that $(\bar{\alpha}^{(i)}, 0, 0)$ is a simple root of $\mathfrak{g}$ comes almost immediately from the definition. To build higher roots we need to be able to increase the grade, i.e. add on multiples of δ . However δ is not itself a simple root since for any $\bar{\alpha} \in \bar{\Phi}_+$ we have

$$(\delta - \bar{\alpha}) + \bar{\alpha} = \delta$$

and so it can be written as a sum of other positive roots. This problem is fixed by the existence of a highest root for $\bar{\mathfrak{g}}$, so that $\delta - \bar{\theta}$ is indeed simple.

Again to define a scalar product on the root space, we first define a scalar product on $\mathfrak{g}$, and then the restriction to the Cartan subalgebra gives a natural scalar product on the root space.

Definition 4.6. The Killing form κ is the unique (up to normalization) real, symmetric, and associative bilinear form on $\mathfrak{g}$.

While we will omit the argument here, it is not hard to show that

$$\begin{aligned}\kappa(T_m^a, T_m^b) &= \delta_{m, -n} \bar{\kappa}(T^a, T^b) \\ \kappa(D, K) &= 1 \\ \kappa(T_m^a, K) &= T(K, K) = T(K, D) = T(D, D) = 0\end{aligned}$$

Under the right normalization and correct basis, the restriction to the Cartan subalgebra is given simply by the matrix

$$\begin{pmatrix} I_{r \times r} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

where $I_{r \times r}$ is the r by r identity matrix. Thus the scalar product of two roots $(\bar{\alpha}, 0, n)$ and $(\bar{\beta}, 0, m)$ is simply the usual inner product $(\bar{\alpha}, \bar{\beta})$. In general, the inner product between two tuples $\lambda = (\bar{\lambda}, k, n)$ and $\lambda' = (\bar{\lambda}', k', n')$ is

$$(\lambda, \lambda') = (\bar{\lambda}, \bar{\lambda}') + kn' + k'n.$$

Unlike in the simple case, this scalar product has some obvious degeneracy, which leads to one of the first major differences between simple and affine Lie algebras.

Definition 4.7. For an affine Lie algebra $\mathfrak{g}$, we separate the roots into two categories. Let $\alpha \in \Phi$. If $(\alpha, \alpha) > 0$ we say that α is a *real root*. If $(\alpha, \alpha) = 0$ then we say α is an *imaginary root*.

Equivalently α is real if it is of the form $(\bar{\alpha}, 0, n)$ while it is imaginary if it is of the form $(0, 0, n)$.

From this second characterization it is clear that all the simple roots are real.

Definition 4.8. To simplify notation, we introduce the *coroots*, which are defined by

$$\alpha^\vee = \frac{2}{(\alpha, \alpha)} \alpha \quad \alpha \in \Phi.$$

The affine Cartan matrix is the $r + 1$ by $r + 1$ matrix with entries

$$A^{ij} = (\alpha^{(i)}, \alpha^{(j)\vee}) \quad i, j = 0, 1, \dots, r.$$

This essentially concludes our discussion of the root system of affine Lie algebras. However it should be noted that the similarities between the root system of affine Lie algebras and semisimple Lie algebras is not coincidental. It turns out that semisimple Lie algebras can be precisely characterized by their Cartan matrix, and that any matrix which satisfies a list of properties is the Cartan matrix for some semisimple Lie algebra. Affine Lie algebras (or more generally, Kac-Moody algebras) are obtained by simply loosening these requirements on the Cartan matrix.

Definition 4.9. *Kac-Moody algebras* are Lie algebras obtained starting from a matrix A which satisfies all the requirements for the Cartan matrix of a semisimple Lie algebra, except we don't require that $\det A > 0$ (see [Theorem 3.18](#)).

In a sense, affine Lie algebras are in between semisimple Lie algebras and this more general class:

Proposition 4.10. *Let A be the $n \times n$ Cartan matrix of a Kac-Moody algebra. Then the algebra it corresponds to is an affine Lie algebra if and only if*

$$\det A_{\{i\}} > 0 \quad i = 1, 2, \dots, n$$

where $A_{\{i\}}$ is the minor matrix without the i^{th} row and column.

Note that this process of construct affine Lie algebras creates two types generally called *twisted* and *untwisted*. Our loop algebra construction only created the untwisted ones, which are the only ones we will discuss here.

Let us consider what we have seen so far for the affine Lie algebra built out of the simple Lie algebra $\mathfrak{sl}(2)$. Going forward we will let $A_1 = \mathfrak{sl}(2)$ and denote the affine Lie algebra built out of it by $A_1^{(1)}$. Recall that the basis for the Cartan subalgebra of A_1 was one dimensional, with generator H . There were then only two roots $\alpha = (+2)$ and $-\alpha = (-2)$, so that the set of positive roots, the set of simple roots, and the highest root all coincide. To get the affine Lie algebra $A_1^{(1)}$ we add a grading (i.e. form the loop algebra) and then extend with K and D . Concretely, our brackets become

$$\begin{aligned} [H_n, H_m] &= \delta_{n-m,0} K \\ [H_n, E_m^\pm] &= \pm 2 E_{n+m}^\pm \\ [K, \cdot] &= 0 \\ [D, H_n] &= n H_n \\ [D, E_n^\pm] &= n E_n^\pm. \end{aligned}$$

Looking at the brackets with H_n for $n \neq 0$ or $E_n^\pm$ on the right we get roots of the form $(0, 0, n)$ and $(\pm 2, 0, n)$. Following [Definition 4.4](#) the positive roots are

$$\{(2, 0, n) : n \geq 0\} \cup \{(a, 0, n) : n > 0, a \in \{-2, 0, 2\}\}.$$

So, unlike in the simple case, we have two simple roots $\alpha_{(0)} = (-2, 0, 1)$ and $\alpha_{(1)} = (2, 0, 0)$, and the corresponding coroots $\alpha_{(0)}^\vee = (1, 0, 0)$ and $\alpha_{(1)}^\vee = (-1, 0, \frac{1}{2})$. The Cartan matrix is therefore

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}.$$

The principal minor matrices are both (2), which does indeed have determinant greater than 0 (but the determinant of the overall matrix is 0, so indeed $A_1^{(1)}$ is not semisimple). Notice that this minor is in fact the Cartan matrix of A_1 .

5. MODULES OVER AFFINE LIE ALGEBRAS

5.1. Modules. In the case of simple Lie algebras, all finite dimensional irreducible modules were highest weight modules, in the sense of [Theorem 3.22](#). In the case of affine Lie algebras, all finite dimensional modules are essentially trivial, so rather than deduce that our modules of interest have a highest weight, we simply make it a requirement. Throughout this section, let $\mathfrak{g}$ be an affine Lie algebra with all the notation from the previous section. Also note that we will use the words representation and module interchangeably.

Definition 5.1. Let R be a $\mathfrak{g}$ -module, let $v \in R$ and suppose that the Cartan subalgebra $\mathfrak{g}_0$ acts diagonally on v . That is there exist $\lambda_1, \dots, \lambda_r, k, n \in \mathbb{C}$ such that

$$\begin{aligned} R_\Lambda(H_0^i) \cdot v &= \lambda_i v \\ R_\Lambda(K) \cdot v &= kv \\ R_\Lambda(D) \cdot v &= nv. \end{aligned}$$

Let $\bar{\lambda} = (\lambda_1, \dots, \lambda_r) \in \mathbb{C}^r$. Then the tuple $\lambda = (\bar{\lambda}, k, n)$ is called a *weight* of $\mathfrak{g}$ for R .

So that, as before, the roots are the weights for the adjoint representation. Note that we really only care about weights of certain modules (namely integrable highest weight modules), so once they have been introduced it will always be assumed that the word weight refers to a weight in such a module.

Definition 5.2. Let $\Lambda = (\bar{\Lambda}, k, n_0)$, where $\bar{\Lambda} \in \mathbb{C}^r$ and $k, n \in \mathbb{C}$. A *highest weight module* R_Λ is a $\mathfrak{g}$ -module satisfying the following properties:

- (1) There exists a unique (up to scalar multiplication) non-zero vector v_Λ such that v_Λ has weight Λ and if T_n^a is a generator not in the Cartan subalgebra corresponding to a positive root (see [Definition 4.3](#)), then

$$R_\Lambda(T_m^a) \cdot v_\Lambda = 0.$$

- (2) All other vectors v in the module can be written (up to a scalar multiple)

$$(5.3) \quad v = R_\Lambda(T_{-n_1}^{a_1}) \dots R_\Lambda(T_{-n_m}^{a_m}) \cdot v_\Lambda$$

where each $T_{-n_i}^{a_i}$ is a generator of $\mathfrak{g}$ corresponding to a negative root.

We are as usual most interested in irreducible representations, however we still give a name to the simplest highest weight representation, even though it is certainly not irreducible:

Definition 5.4. The *Verma module* V_Λ is the highest weight module obtained by taking all the expressions of the form (5.3) to be formally linearly independent.

Suppose that T_n^a corresponds to the negative root $\alpha \in \Phi_-$, and v is a vector with weight λ . It is easy to show that $R_\Lambda(T_n^a) \cdot v$ is a vector of weight $\lambda + \alpha$. Thus in any highest weight representation R_Λ , the Cartan subalgebra acts diagonally, i.e. we can write

$$R_\Lambda = \bigoplus R_{(\lambda)}$$

where $R_{(\lambda)}$ is the vector subspace consisting of vectors with weight λ .

Now our goal is to constrain the highest weight Λ in order to create a “well-behaved” module. We first note that the D eigenvalue, n_0 , is essentially irrelevant, since if we replace D by $\tilde{D} = D - n_0 \frac{K}{k}$ (assuming $k \neq 0$, which is true in all non-trivial cases) then $\tilde{D}$ has all the same commutators as D and eigenvalue 0. Thus we will assume that $n_0 = 0$.

Observation 5.5. Since K is central, if $R_\Lambda(K) \cdot v_\Lambda = kv_\Lambda$ then $R_\Lambda(K) \cdot v = kv$ for all v in the module. The normalized value

$$k^\vee := k \cdot \frac{2}{(\bar{\theta}, \bar{\theta})}.$$

is called the *level* of R_Λ .

From the commutation relations of D we get that the D eigenvalue of a vector $v = R_\Lambda(T_{-n_1}^{a_1}) \dots R_\Lambda(T_{-n_m}^{a_m}) \cdot v_\Lambda$ is $-(\sum_j n_j)$, which is called the *grade* of v .

For each real root $\alpha = (\bar{\alpha}, 0, n)$ we can find a corresponding $\mathfrak{sl}(2)$ subalgebra. The generators are $E^\alpha = E_n^{\bar{\alpha}}, E^{-\alpha} = E_{-n}^{-\bar{\alpha}}$ and

$$\begin{aligned} H^\alpha &= \frac{2}{(\alpha, \alpha)} [E^\alpha, E^{-\alpha}] \\ &= \frac{2}{(\alpha, \alpha)} \left(([E^{\bar{\alpha}}, E^{-\bar{\alpha}}]_{\bar{\mathfrak{g}}} \otimes t^0) + nK \right) \\ &= \frac{2}{(\alpha, \alpha)} \left(\sum_{i=1}^r \bar{\alpha}_i H_0^i + nK \right). \end{aligned}$$

We leave it to the reader to check that these generate an $\mathfrak{sl}(2)$ subalgebra, and we note that the set of these triples for all $\alpha \in \Phi$ form a basis for $\bar{\mathfrak{g}}$, called the *Chevalley basis*. For any weight $\lambda = (\bar{\lambda}, k, m)$ for the module R_Λ we have

$$R_\Lambda(H^\alpha) \cdot v_\lambda = \frac{2}{(\alpha, \alpha)} \left(\sum_{i=1}^r \bar{\alpha}_i \bar{\lambda}^i + nk \right) v_\lambda = (\lambda, \alpha^\vee) v_\lambda.$$

Recall from the simple case that a module was finite dimensional if and only if the H^α eigenvalues were integers for any weight λ , and a non-negative integer for the highest weight. In our case, we do not want ensure the module is finite dimensional (all finite dimensional irreducible modules for an affine Lie algebra are trivial) but simply making this a requirement ensures our modules are well-behaved:

Definition 5.6. A highest weight module R_Λ is *integrable* if for any real root α we have

$$(\lambda, \alpha^\vee) \in \mathbb{Z} \quad \text{and} \quad (\Lambda, \alpha^\vee) \in \mathbb{Z}_{\geq 0}.$$

Proposition 5.7. A module R_Λ is integrable if and only if

$$(\Lambda, \alpha^{(i)\vee}) \in \mathbb{Z}_{\geq 0}$$

for $i = 0, 1, \dots, r$ and $\alpha^{(i)}$ the simple roots of $\bar{\mathfrak{g}}$.

Proof. This comes directly from the bilinearity of the Killing form and the fact that all weights in a highest weight module can be found by subtracting positive roots from the highest weight. $\square$

For example, consider the simple root $\alpha^{(0)} = (-\bar{\theta}, 0, 1)$. Then we have

$$\left((-\bar{\theta}^\vee, 0, 2/(\bar{\theta}, \bar{\theta})), (\bar{\Lambda}, k, 0) \right) = k^\vee - (\bar{\Lambda}, \bar{\theta}^\vee) \in \mathbb{Z}_{\geq 0}$$

and so, since $(\bar{\Lambda}, \bar{\theta}^\vee) = (\bar{\Lambda}, \theta^\vee)$ is an integer by assumption, we conclude that the level $k^\vee$ is always an integer and

$$(\bar{\Lambda}, \bar{\theta}^\vee) \leq k^\vee$$

which is a useful criterion for constraining the number of highest weights at a given level (indeed it is not too hard to show that there are only finitely many choices at each level). We introduce the following definition from semisimple Lie algebras:

Definition 5.8. Let θ be the highest root of a semisimple Lie algebra $\mathfrak{g}$. By the definition of simple roots (and so simple coroots) we can write

$$\theta^\vee = \sum_{i=1}^r a_i^\vee \alpha^{(i)\vee}$$

where the coefficients $a_i^\vee$ are non-negative integers called the *dual Coxeter labels* of $\mathfrak{g}$. The *dual Coxeter number* is then

$$g^\vee = 1 + \sum_{i=1}^r a_i^\vee.$$

Back to affine Lie algebras, we have

Definition 5.9. The *fundamental weights* are the weights $\Lambda_{(i)}$ such that

$$(\Lambda_{(i)}, \alpha^{(j)\vee}) = \delta_{ij}.$$

The *Weyl vector* ρ is then defined by

$$\sum_{i=0}^r \Lambda_{(i)}.$$

Note that is equivalent to our definition (3.25) in the simple case if we define the fundamental weights analogously.

Proposition 5.10. *The fundamental weights are*

$$\Lambda_{(0)} = (0, \tfrac{1}{2}(\bar{\theta}, \bar{\theta}), 0), \quad \Lambda_{(i)} = (\bar{\Lambda}_{(i)}, \tfrac{1}{2}(\bar{\theta}, \bar{\theta})a_i^\vee, 0)$$

where $\bar{\Lambda}_{(i)}$ are the fundamental weights of the corresponding simple Lie algebra. The Weyl vector is then

$$\rho = (\bar{\rho}, \tfrac{1}{2}(\bar{\theta}, \bar{\theta})g^\vee, 0).$$

We leave it to the reader to check that $(\Lambda_{(i)}, \alpha^{(j)\vee}) = \delta_{ij}$.

Again we will do some explicit working out for $A_1^{(1)}$, although we don't have enough structure to really understand the integrable highest weight modules of $A_1^{(1)}$. Suppose R_Λ is such a module, with highest weight $\Lambda = (\bar{\Lambda}, k, 0)$. Then Proposition 5.7 tells us

$$(\Lambda, \alpha_{(0)}^\vee) = -2\bar{\Lambda} + \tfrac{1}{2}k \in \mathbb{Z}_{\geq 0} \quad \text{and} \quad (\Lambda, \alpha_{(1)}^\vee) = \bar{\Lambda} \in \mathbb{Z}_{\geq 0}.$$

Combining these criteria we indeed get that k must be a positive even integer, and then $k^\vee = \frac{2}{(\bar{\theta}, \bar{\theta})}k = \tfrac{1}{2}k$, so $k^\vee$ can be any positive integer. At level $k^\vee = 1$, we have

$$k = \frac{(\bar{\theta}, \bar{\theta})}{2}k^\vee = 2.$$

Thus we have $-2\bar{\Lambda} + 1 \in \mathbb{Z}_{\geq 0}$ so $\bar{\Lambda}$ must be 0. Thus there is a unique highest weight module of level 1, with highest weight $(0, 2, 0)$. Since the only simple root is the highest root, $g^\vee = 2$ and the Weyl vector is $\rho = (1, 4, 0)$

5.2. Characters and the Kac-Weyl Formula. Now we turn our attention to the character of these integrable highest weight modules. The character of some $H \in \mathfrak{g}_0$ can still be thought of as the (now infinite-dimensional) trace of $e^{R_\Lambda(H)}$, however we just define it directly as the sum:

Definition 5.11. If R_Λ is an integrable highest weight module, then the character of R_Λ is given by

$$\chi_\Lambda = \sum_{\lambda} \text{mult}_\Lambda(\lambda) e^\lambda$$

where e^λ is defined in the same way as for the simple case (and again can either be considered with domain $\mathfrak{g}_0$ or $L_w(\mathfrak{g})$ the weight lattice).

Recall from [Observation 5.5](#) that highest weight modules have a natural ‘grading’ by their D eigenvalues, so this infinite dimensional trace evaluated at a multiple of D becomes particularly simple. In particular, multiples of D correspond to the weights $(0, \xi, 0)$, since taking the inner product with $\lambda = (\bar{\lambda}, k, n)$ gives ξn .

Definition 5.12. The *basic specialization* of the character is $\text{Tr}_{R_\Lambda} e^{\xi D}$ or, writing $\xi = -2\pi i \tau$,

$$\chi_\Lambda(0, \xi, 0) = \sum_{n \in \mathbb{Z}_{\geq 0}} d_n q^n$$

where d_n is the dimension of the subspace spanned by vectors of grade n .

Thus the basic specialization is just the generating function for the dimension of the graded subspaces. We will return to this specialization later.

The key difference between the affine case and the simple case is that our sums are now infinite, and so cannot be computed by hand (as we did for A_1 for example). Thus, while for simple Lie algebras the Weyl character formula was computationally and conceptually helpful, for affine Lie algebras the analogous result becomes necessary.

Definition 5.13. For any real root α , the *Weyl reflection* $\sigma_\alpha : L_w \rightarrow L_w$ is given by

$$\sigma_\alpha(\lambda) = \lambda - (\lambda, \alpha^\vee) \alpha.$$

The group W generated by these reflections is called the *affine Weyl group*.

Note that in general $\sigma_\alpha(\lambda)$ is in the weight lattice only if $(\lambda, \alpha^\vee) \in \mathbb{Z}$, so it is very important that our module is integrable. Furthermore, while we chose to give the formula here, it is still the case that σ_α is indeed the reflection at the hyperplane perpendicular to α . As in the simple case, we can define the sign of any $\sigma \in W$ as the unique nontrivial homomorphism from W to $\{\pm 1\}$. Unlike in the simple case, however, the affine Weyl group is infinite.

Theorem 5.14 (Kac-Weyl character formula). *If R_Λ is an integrable highest weight module, then*

$$\chi_\Lambda = \frac{\sum_{\sigma \in W} (\text{sign } \sigma) e^{\sigma(\Lambda + \rho)}}{\sum_{\sigma \in W} (\text{sign } \sigma) e^{\sigma(\rho)}}$$

and the denominator identity holds as well, with a slight modification to account for the fact that roots can now have multiplicity (in particular, imaginary roots have multiplicity r):

$$\sum_{\sigma \in W} (\text{sign } \sigma) e^{\sigma(\rho)} = e^\rho \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{mult}(\alpha)}$$

Note that one has to be careful with this quotient of infinite sums. For example, it is possible that when evaluated individually the numerator and denominator do not converge or both equal 0, etc. However, everything works after canceling out terms and such. This can of course be defined rigorously, but since we are not even proving the Kac-Weyl character formula, we will not explore this here.

Naively this formula tells us that one infinite sum is equal to the quotient to two different infinite sums, and so may not seem that useful for computation. However, we can make use of the structure of the affine Weyl group.

Theorem 5.15. *We have $W = \overline{W} \ltimes T$ where $\overline{W}$ is the Weyl group of $\overline{\mathfrak{g}}$ and T is the group of translations $t_{\overline{\beta}} : L_w \rightarrow L_w$ of the form*

$$t_{\overline{\beta}}(\overline{\lambda}, k, m) = \left(\overline{\lambda} + k\overline{\beta}, k, m - (\overline{\lambda}, \overline{\beta}) - k \frac{(\overline{\beta}, \overline{\beta})}{2} \right)$$

for any $\overline{\beta}$ in the coroots of $\overline{\mathfrak{g}}$, $L^\vee(\overline{\mathfrak{g}})$. In particular, for any $\sigma \in W$ we can write

$$\sigma = \overline{\sigma} \circ t_{\overline{\beta}}$$

for some $\overline{\beta} \in L^\vee(\overline{\mathfrak{g}})$ and some $\overline{\sigma} \in \overline{W}$ (which is taken to act as $\overline{\sigma}$ on the first component of the weight and the identity on the second two components).

Proof. We will focus on showing the second part, as the statement that $W = \overline{W} \ltimes T$ essentially follows and is not particularly important for us. First consider the basic reflection σ_α with $\alpha = (\overline{\alpha}, 0, n)$ acting on $\lambda = (\overline{\lambda}, k, m)$. Note that $(\alpha, \alpha) = (\overline{\alpha}, \overline{\alpha})$.

$$\begin{aligned} \sigma_\alpha(\lambda) &= \lambda - (\lambda, \alpha^\vee) \alpha \\ &= \lambda - \frac{2}{(\alpha, \alpha)} ((\overline{\lambda}, \overline{\alpha}) + kn) \alpha \\ &= \lambda - \left(\overline{\lambda}, \frac{2}{(\overline{\alpha}, \overline{\alpha})} \overline{\alpha} \right) \alpha - kn \frac{2}{(\alpha, \alpha)} \alpha \\ &= \lambda - (\overline{\lambda}, \overline{\alpha}^\vee) \alpha - kn \alpha^\vee \\ &= \left(\overline{\lambda} - (\overline{\lambda}, \overline{\alpha}^\vee) \overline{\alpha} - kn \overline{\alpha}^\vee, k, m - (\overline{\lambda}, \overline{\alpha}^\vee) n - \frac{2}{(\overline{\alpha}, \overline{\alpha})} kn^2 \right) \\ &= \left(\sigma_{\overline{\alpha}}(\overline{\lambda} + nk \overline{\alpha}^\vee), k, m - (\overline{\lambda}, n \overline{\alpha}^\vee) - k \frac{(n \overline{\alpha}^\vee, n \overline{\alpha}^\vee)}{2} \right) \\ &= (\overline{\sigma}_{\overline{\alpha}} \circ t_{n \overline{\alpha}^\vee})(\lambda). \end{aligned}$$

Thus we have the desired fact for all simple reflections. To extend this to all elements of W we will show that $\overline{\sigma}_{\overline{\alpha}} \circ t_{\overline{\beta}} \circ \overline{\sigma}_{\overline{\alpha}}^{-1} = t_{\overline{\sigma}_{\overline{\alpha}}(\overline{\beta})}$. Recall that $\overline{\sigma}_{\overline{\alpha}}$ is its own

inverse. Thus we compute

$$\begin{aligned}
(\bar{\sigma}_{\bar{\alpha}} \circ t_{\bar{\beta}} \circ \bar{\sigma}_{\bar{\alpha}})(\bar{\lambda}, k, m) &= (\bar{\sigma}_{\bar{\alpha}} \circ t_{\bar{\beta}})(\bar{\lambda} - (\bar{\lambda}, \bar{\alpha}^\vee)\bar{\alpha}, k, m) \\
&= \bar{\sigma}_{\bar{\alpha}}\left(\bar{\lambda} - (\bar{\lambda}, \bar{\alpha}^\vee)\bar{\alpha} + k\bar{\beta}, k, m - (\bar{\lambda} - (\bar{\lambda}, \bar{\alpha}^\vee)\bar{\alpha}, \bar{\beta}) - \frac{k}{2}(\bar{\beta}, \bar{\beta})\right) \\
&= \left(\bar{\lambda} - (\bar{\lambda}, \bar{\alpha}^\vee)\bar{\alpha} + k\bar{\beta} - (\bar{\lambda}, \bar{\alpha}^\vee)\bar{\alpha} + 2(\bar{\lambda}, \bar{\alpha}^\vee)\bar{\alpha} - k(\bar{\beta}, \bar{\alpha}^\vee)\bar{\alpha}, \right. \\
&\quad \left. k, m - (\bar{\lambda} - (\bar{\lambda}, \bar{\alpha}^\vee)\bar{\alpha}, \bar{\beta}) - \frac{k}{2}(\bar{\beta}, \bar{\beta})\right) \\
&= \left(\bar{\lambda} + k(\bar{\beta} - (\bar{\beta}, \bar{\alpha}^\vee)\bar{\alpha}), k, m - (\bar{\lambda} - (\bar{\lambda}, \bar{\alpha}^\vee)\bar{\alpha}, \bar{\beta}) - \frac{k}{2}(\bar{\beta}, \bar{\beta})\right) \\
&= t_{\bar{\sigma}_{\bar{\alpha}}(\bar{\beta})}(\bar{\lambda}, k, m)
\end{aligned}$$

Similar computations show that the map $\bar{\beta} \mapsto t_{\bar{\beta}}$ is actually a group isomorphism from $L^\vee(\bar{\mathfrak{g}})$ to T in that

$$t_{\bar{\beta}_1} \circ t_{\bar{\beta}_2} = t_{\bar{\beta}_1 + \bar{\beta}_2}.$$

Note that this justifies our referring to the elements of T as translations.

Now consider the product of two basic reflections $\sigma_{\alpha_1}, \sigma_{\alpha_2} \in W$. By the above we can write $\sigma_{\alpha_1} = \bar{\sigma}_{\bar{\alpha}_1} \circ t_{\bar{\beta}_1}$ for some $\bar{\beta}_1 \in L^\vee(\bar{\mathfrak{g}})$, and similarly for σ_{α_2} . Thus

$$\begin{aligned}
\sigma_{\alpha_1} \circ \sigma_{\alpha_2} &= (\bar{\sigma}_{\bar{\alpha}_1} \circ t_{\bar{\beta}_1}) \circ (\bar{\sigma}_{\bar{\alpha}_2} \circ t_{\bar{\beta}_2}) \\
&= \bar{\sigma}_{\bar{\alpha}_1} \circ \bar{\sigma}_{\bar{\alpha}_2} \circ t_{\bar{\sigma}_{\bar{\alpha}_2}(\bar{\beta}_1)} \circ t_{\bar{\beta}_1} \\
&= (\bar{\sigma}_{\bar{\alpha}_1} \circ \bar{\sigma}_{\bar{\alpha}_2}) \circ t_{\bar{\sigma}_{\bar{\alpha}_2}(\bar{\beta}_1) + \bar{\beta}_1}
\end{aligned}$$

which is indeed of the form $\bar{\sigma} \circ t_{\bar{\beta}}$. Since the reflections σ_α generate W , this result extends inductively to all of W . $\square$

Taking for example the denominator of the Weyl-Kac character formula we can now write

$$(5.16) \quad \sum_{\sigma \in W} (\text{sign } \sigma) e^{\sigma(\rho)} = \sum_{\bar{\sigma} \in \bar{W}} \left[(\text{sign } \bar{\sigma}) \sum_{\bar{\beta} \in L^\vee(\bar{\mathfrak{g}})} e^{t_{\bar{\beta}} \bar{\sigma}(\rho)} \right]$$

so we have a finite sum of sums over a lattice—something that seems much nicer to work with.

5.3. Modular Properties of Characters. Now that we are taking a sum over a lattice it is perhaps not surprising that we are very close to the theta functions from [Section 2](#). To make this precise, we will define a new Θ function.

Definition 5.17. Given a coroot lattice $L^\vee$ for some simple Lie algebra, some $\bar{\mu} \in L^\vee$ and a level $k^\vee$ let

$$\Theta_{L^\vee, \bar{\mu}, k^\vee}(\zeta, \tau, t) = e^{2\pi i k^\vee t} \sum_{\bar{\beta} \in L^\vee + k^{-1}\bar{\mu}} e^{2\pi i k^\vee \langle \bar{\beta}, \zeta \rangle} q^{k^\vee \langle \bar{\beta}, \bar{\beta} \rangle / 2}$$

This new theta function essentially combines our two theta functions from [Definition 2.6](#). Indeed we have

$$\Theta_{L^\vee, \bar{\mu}, k^\vee}(0, \tau, 0) = \sum_{\bar{\beta} \in L^\vee + k^{-1}\bar{\mu}} q^{k^\vee \langle \bar{\beta}, \bar{\beta} \rangle / 2}$$

which just looks like a sum over a lattice shifted by some scaled vector in that lattice. In fact, the only reason we did not define this more general theta function in that section is that $k^\vee$ and k are related by a factor of $2/(\bar{\theta}, \bar{\theta})$ which depends on the Lie algebra.³ In particular

$$\Theta_{L^\vee, 0, 1}(0, \tau, 0) = \Theta_{L^\vee}(\tau)$$

where the right hand side is just the normal lattice theta function. On the other hand, since A_1 is the unique simple Lie algebra with coroot lattice $\mathbb{Z}$, if $L^\vee = \mathbb{Z}$ we have $k = 2k^\vee$ and so

$$(5.18) \quad \Theta_{\mathbb{Z}, m, k^\vee}(\zeta, \tau, t) = e^{2\pi i k^\vee t} \sum_{\bar{\beta} \in \mathbb{Z} + \frac{m}{2k^\vee}} e^{2\pi i k^\vee (\bar{\beta}, \zeta)} q^{k^\vee (\bar{\beta}, \bar{\beta})/2} = \Theta_{m, k^\vee}(\zeta, \frac{\tau}{2}, t)$$

(we will explore this further in the next section and see why the factor of $\frac{\tau}{2}$ comes up). The other reason these theta functions were not introduced earlier is that they are not modular forms in the traditional sense. However, given their close relation to the theta functions from [Section 2](#) we should not be surprised that they have nice transformation properties under the action of the modular group. However, these transformations are still somewhat messy, so we will focus on reducing them to the simpler cases rather than giving these general properties.

Now we go back to the characters of affine Lie algebras.

Proposition 5.19. *Given a weight $\lambda = (\bar{\lambda}, \xi, \nu)$ consider the change of variables $(\bar{\lambda}, \xi, \nu) = \frac{4\pi i}{(\bar{\theta}, \bar{\theta})}(\zeta, -\tau, t)$. Then*

$$\chi_\Lambda(\lambda) = e^{s_\Lambda \delta} \frac{\sum_{\bar{\sigma} \in \bar{W}} (\text{sign } \bar{\sigma}) \Theta_{L^\vee, \bar{\sigma}(\Lambda + \rho), g^\vee + k^\vee}(\zeta, \tau, t)}{\sum_{\bar{\sigma} \in \bar{W}} (\text{sign } \bar{\sigma}) \Theta_{L^\vee, \bar{\sigma}(\rho), g^\vee}(\zeta, \tau, t)}$$

where

$$s_\Lambda = \frac{1}{(\bar{\theta}, \bar{\theta})} \left(\frac{(\Lambda + \rho, \Lambda + \rho)}{k^\vee + g^\vee} - \frac{(\rho, \rho)}{g^\vee} \right).$$

which is called the modular anomaly of R_Λ .

Proof. The proof is essentially just computation. Substituting $(\zeta, \tau, t) = \frac{(\bar{\theta}, \bar{\theta})}{4\pi i}(\bar{\lambda}, -\xi, \nu)$ into [Definition 5.17](#) we get

$$\begin{aligned} \Theta_{L^\vee, \bar{\mu}, \kappa^\vee}(\bar{\lambda}, \xi, \nu) &= e^{\kappa^\vee \frac{(\bar{\theta}, \bar{\theta})}{2} \nu} \sum_{\bar{\beta} \in L^\vee + \kappa^{-1} \bar{\mu}} e^{\kappa^\vee \frac{(\bar{\theta}, \bar{\theta})}{2} ((\bar{\beta}, \bar{\lambda}) - \frac{1}{2}(\bar{\beta}, \bar{\beta}) \xi)} \\ &= \left(e^{(0, \kappa, 0)} \sum_{\bar{\beta} \in L^\vee + \kappa^{-1} \bar{\mu}} e^{\kappa (\bar{\beta} - \frac{1}{2}(\bar{\beta}, \bar{\beta}) \delta)} \right) (\bar{\lambda}, \xi, \nu) \end{aligned}$$

³Sometimes the inner product is normalized such that $(\bar{\theta}, \bar{\theta}) = 2$, in which case this value does not depend on the Lie algebra but the inner product itself does.

Pulling the outside term into the sum and moving the shift from the lattice to the exponent we get

$$\begin{aligned}\Theta_{L^\vee, \bar{\mu}, \kappa^\vee} &= \sum_{\bar{\beta} \in L^\vee} e^{\left(\bar{\mu} + \kappa \bar{\beta}, \kappa, -\frac{1}{2\kappa}(\bar{\mu} + \kappa \bar{\beta}, \bar{\mu} + \kappa \bar{\beta})\right)} \\ &= e^{-\frac{1}{2\kappa}(\mu, \mu)\delta} \sum_{\bar{\beta} \in L^\vee} e^{t_{\bar{\beta}}(\mu)}\end{aligned}$$

where $\mu = (\bar{\mu}, \kappa, 0)$. We will now plug this into (5.16) for the denominator of the Weyl-Kac formula. Recall that $\rho = (\bar{\rho}, \frac{1}{2}(\bar{\theta}, \bar{\theta})g^\vee, 0)$. Thus we use $\kappa = \frac{1}{2}(\bar{\theta}, \bar{\theta})g^\vee$ and $\bar{\sigma}(\rho) = (\bar{\sigma}(\bar{\rho}), \kappa, 0)$ and so

$$\sum_{\sigma \in W} (\text{sign } \sigma) e^{\sigma(\rho)} = \sum_{\bar{\sigma} \in \bar{W}} (\text{sign } \bar{\sigma}) e^{(\bar{\sigma}(\rho), \bar{\sigma}(\rho))\delta / ((\bar{\theta}, \bar{\theta})g^\vee)} \Theta_{L^\vee, \bar{\sigma}(\rho), g^\vee}$$

Note that the simple Weyl group leaves the inner product invariant. Thus $(\bar{\sigma}(\rho), \bar{\sigma}(\rho)) = (\rho, \rho)$ and we can pull this factor out of the sum to get:

$$e^{(\rho, \rho)\delta / ((\bar{\theta}, \bar{\theta})g^\vee)} \sum_{\bar{\sigma} \in \bar{W}} (\text{sign } \bar{\sigma}) \Theta_{L^\vee, \bar{\sigma}(\rho), g^\vee}.$$

For the numerator we have $\Lambda + \rho = (\bar{\Lambda} + \bar{\rho}, \frac{1}{2}(\bar{\theta}, \bar{\theta})g^\vee + k, 0)$ (where k is now the level of the representation) so by essentially the same manipulations we get

$$\sum_{\sigma \in W} (\text{sign } \sigma) e^{\sigma(\Lambda + \rho)} = e^{(\Lambda + \rho, \Lambda + \rho)\delta / ((\bar{\theta}, \bar{\theta})g^\vee + 2k)} \sum_{\bar{\sigma} \in \bar{W}} (\text{sign } \bar{\sigma}) \Theta_{L^\vee, \bar{\sigma}(\Lambda + \rho), g^\vee + k^\vee}.$$

Putting these together we indeed get the desired result. $\square$

Thus we find that the character of any integrable highest weight representation of any affine Lie algebra is a simple ratio of finite sums of theta functions, with some prefactor. In fact, for simplicity and modularity purposes we make a new definition.

Definition 5.20. The *modified character* $\tilde{\chi}_\Lambda$ (which some authors simply call the character) is the character without the $e^{s_\Lambda \delta}$ factor, i.e.

$$\begin{aligned}\tilde{\chi}_\Lambda &= e^{-s_\Lambda \delta} \chi_\Lambda \\ &= \frac{\sum_{\bar{\sigma} \in \bar{W}} (\text{sign } \bar{\sigma}) \Theta_{L^\vee, \bar{\sigma}(\Lambda + \rho), g^\vee + k^\vee}(\zeta, \tau, t)}{\sum_{\bar{\sigma} \in \bar{W}} (\text{sign } \bar{\sigma}) \Theta_{L^\vee, \bar{\sigma}(\rho), g^\vee}(\zeta, \tau, t)}\end{aligned}$$

5.4. Characters of $A_1^{(1)}$. Now we will again focus on $A_1^{(1)}$, although we will be able to go a little further than we were for a generic affine Lie algebra. Our particular goal will be to show that the basic specialization of the modified character of the level one representation of $A_1^{(1)}$ is simply

$$\tilde{\chi}_\Lambda = \frac{\Theta(\tau)}{\eta(\tau)}$$

where

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n)$$

is the Dedekind eta function (which is a weight $1/2$ modular form). The fact that this somewhat miraculous formula arises for the simplest affine Lie algebra $A_1^{(1)}$ is our first concrete example of moonshine (of a certain variety).

Let us establish everything we need to know about $A_1^{(1)}$ at level one in order to plug into [Proposition 5.19](#). We have $\bar{\theta} = 2$, $\Lambda = (0, 2, 0)$, $g^\vee = 2$, $\rho = (1, 4, 0)$, and since we are at level 1, $k^\vee = 1$. Thus the modular anomaly is

$$\begin{aligned} s_\Lambda &= \frac{1}{(\bar{\theta}, \bar{\theta})} \left(\frac{(\Lambda + \rho, \Lambda + \rho)}{k^\vee + g^\vee} - \frac{(\rho, \rho)}{g^\vee} \right) \\ &= \frac{1}{4} \left(\frac{1}{1+2} - \frac{1}{2} \right) \\ &= -\frac{1}{24} \end{aligned}$$

The coroot lattice is just $\mathbb{Z}$ (under our normalization) and the simple Weyl group is just $\{\pm 1\}$. Thus the modified character is

$$\tilde{\chi}_\Lambda = \frac{\Theta_{\mathbb{Z}, 1, 3} - \Theta_{\mathbb{Z}, -1, 3}}{\Theta_{\mathbb{Z}, 1, 2} - \Theta_{\mathbb{Z}, -1, 2}}$$

Now we would like to move to the basic specialization as in [Definition 5.12](#). Recall that we set $\xi = -2\pi i\tau$ to get that the basic specialization is the generating function of the grade. This is slightly different then our previous change of variables $(\bar{\lambda}, -\xi, \nu) = (4\pi i/(\bar{\theta}, \bar{\theta}))(\zeta, \tau, t) = \pi i(\bar{\lambda}, \xi, \nu)$, in particular they are off by a factor of 2.⁴ Thus for the basic specialization, which we will still denote $\tilde{\chi}_\Lambda(\tau)$, we use $(\bar{\lambda}, -\xi, \nu) = (4\pi i/(\bar{\theta}, \bar{\theta}))(\zeta, 2\tau, t)$. Applying [\(5.18\)](#) we then get

$$\tilde{\chi}_\Lambda(\tau) = \frac{\Theta_{1,3}(0, \tau, 0) - \Theta_{-1,3}(0, \tau, 0)}{\Theta_{1,2}(0, \tau, 0) - \Theta_{-1,2}(0, \tau, 0)}$$

Theorem 5.21. $\tilde{\chi}_\Lambda(\tau) = \frac{\Theta(\tau)}{\eta(\tau)}$.

Proof. We will use [Proposition 2.9](#) twice to expand

$$\Theta_{0,1}(\tau)(\Theta_{1,2} - \Theta_{-1,2})$$

For both expansions we have $m + m' = 1 + 2 = 3$ and so j ranges over $\mathbb{Z}/3\mathbb{Z}$. Noting that $\Theta_{0,3} = \Theta_{\pm 3,3}$, we get

$$\begin{aligned} &\left(\sum_{k \in \mathbb{Z} - \frac{5}{12}} q^{6k^2} \right) \Theta_{-1,3} + \left(\sum_{k \in \mathbb{Z} - \frac{1}{12}} q^{6k^2} \right) \Theta_{1,3} + \left(\sum_{k \in \mathbb{Z} + \frac{3}{12}} q^{6k^2} \right) \Theta_{0,3} \\ &\quad - \left(\sum_{k \in \mathbb{Z} - \frac{3}{12}} q^{6k^2} \right) \Theta_{0,3} + \left(\sum_{k \in \mathbb{Z} + \frac{1}{12}} q^{6k^2} \right) \Theta_{-1,3} + \left(\sum_{k \in \mathbb{Z} + \frac{5}{12}} q^{6k^2} \right) \Theta_{1,3}. \end{aligned}$$

Since all our functions are functions of k^2 , we can exchange k and $-k$ in any of these sums. This then has the affect of changing the sign of the shift to the $\mathbb{Z}$ lattice. Thus the coefficient on $\Theta_{0,3}$ is

$$\sum_{k \in \mathbb{Z} + \frac{3}{12}} q^{6k^2} - \sum_{k \in \mathbb{Z} - \frac{3}{12}} q^{6k^2} = 0$$

⁴They are the same under the convention $(\bar{\theta}, \bar{\theta}) = 2$ which we are not using.

leaving us with

$$\begin{aligned} & \left(\sum_{k \in \mathbb{Z} - \frac{1}{12}} q^{6k^2} - \sum_{k \in \mathbb{Z} + \frac{5}{12}} q^{6k^2} \right) \Theta_{1,3} - \left(\sum_{k \in \mathbb{Z} + \frac{1}{12}} q^{6k^2} - \sum_{k \in \mathbb{Z} - \frac{5}{12}} q^{6k^2} \right) \Theta_{-1,3} \\ &= \left(\sum_{k \in \mathbb{Z} - \frac{1}{12}} q^{6k^2} - \sum_{k \in \mathbb{Z} + \frac{5}{12}} q^{6k^2} \right) (\Theta_{1,3} - \Theta_{-1,3}) \end{aligned}$$

and so we focus on this coefficient. We notice that $-\frac{1}{12}$ and $\frac{5}{12}$ are $\frac{1}{2}$ apart, so

$$\begin{aligned} \sum_{k \in \mathbb{Z} - \frac{1}{12}} q^{6k^2} - \sum_{k \in \mathbb{Z} + \frac{5}{12}} q^{6k^2} &= \sum_{k \in \frac{1}{2}\mathbb{Z} - \frac{1}{12}} (-1)^{2(k + \frac{1}{12})} q^{6k^2} \\ &= \sum_{k \in \mathbb{Z}} (-1)^k q^{6(\frac{k}{2} - \frac{1}{12})^2} \\ &= q^{1/24} \sum_{k \in \mathbb{Z}} (-1)^k q^{(3k^2 - k)/2} \end{aligned}$$

which is precisely $\eta(\tau)$ by Euler's famous pentagonal number theorem. Thus indeed we find that

$$\tilde{\chi}_\Lambda(\tau) = \frac{\Theta_{1,3}(\tau) - \Theta_{-1,3}(\tau)}{\Theta_{1,2}(\tau) - \Theta_{-1,2}(\tau)} = \frac{\Theta_{0,1}(\tau)}{\eta(\tau)} = \frac{\Theta(\tau)}{\eta(\tau)}$$

as desired. $\square$

Notice that it was not particularly easy to get from [Proposition 5.19](#) to this result, and it is not obvious that it generalizes. We will now turn to study vertex algebras. Once we have built up this theory enough, a generalized version of this theorem almost will pop out almost for free in [Theorem 7.5](#).

6. VERTEX ALGEBRAS

6.1. Motivation. Let f be an analytic function on $\mathbb{R}$. Here's a suspicious-looking equation

$$\left(e^{z \frac{d}{dx}} f \right) (x) = f(x + z)$$

It can be “verified” by recognizing that the left side is just the Taylor series of f .

$$\sum_n \frac{z^n}{n!} \frac{d^n}{dx^n} f$$

An expression like this is called a *field*

Definition 6.1. Let V be a vector space over $\mathbb{C}$. A *field* on V is a formal power series

$$A(z) = \sum_{j \in \mathbb{Z}} A_j z^{-j} \in (\text{End } V)[[z^{\pm 1}]]$$

such that for all $v \in V$, $A_j v = 0$ for j sufficiently large.

In our case, since our expression doesn't have negative powers of z , we have for every v that $A_j v = 0$ when $j \geq 1$, where the endomorphism is given by multiplication by the function.

$$\frac{1}{(-j)!} \frac{d^{(-j)} f}{dx^{(-j)}}$$

We can rigorously work with power series like these by defining a *vertex algebra*.

Definition 6.2. A *vertex algebra* is a vector space V with a *vacuum vector* $|0\rangle$, *translation operator* $T : V \rightarrow V$, and an operation

$$Y(\cdot, z) : V \rightarrow (\text{End } V)[[z^{\pm 1}]]$$

$$Y(A, z) = \sum_{n \in \mathbb{Z}} A_{(n)} z^{-n-1}$$

taking a state to a field. These must satisfy the following axioms:

- $Y(|0\rangle, z) = \text{Id}_V$ and $T|0\rangle = 0$. In other words, the vacuum behaves as expected.
- For all $A \in V$, $Y(A, z)|0\rangle$ has no negative powers and $Y(A, z)|0\rangle$ evaluated at $z = 0$ is A . This is the statement of the *state-field correspondence*.
- $[T, Y(A, z)] = \frac{\partial}{\partial z} Y(A, z)$ as fields. This motivates the term “translation operator”, as will be seen.
- The operators satisfy a *locality relation*: for any two fields $Y(A, z)$ and $Y(B, w)$, there exists an N such that

$$(z - w)^N Y(A, z) Y(B, w) = (z - w)^N Y(B, w) Y(A, z)$$

The locality condition is hard to justify but is analogous to the statement that two operators with spacelike separation should commute. (We can't just divide by $(z - w)^N$ since we're working with formal power series.)

Example 6.3. Any commutative algebra with a derivation T is a vertex algebra. For a concrete example, consider real analytic functions on $\mathbb{R}$ with $T = \frac{d}{dx}$. Let $|0\rangle \equiv 1$ and the state-to-field map be given by

$$Y(A, z) = \sum_{n \in \mathbb{Z}} \left(\frac{1}{(-n-1)!} \frac{d^{(-n-1)} A}{dx^{(-n-1)}} \right) z^{-n-1}$$

We can then verify

- $Y(1, z) = 1$ and $\frac{d}{dx}(1) = 0$, so the vacuum axiom is satisfied.
- $Y(A, 0) = A$, verifying the state-field correspondence.
- Using our “shift operator” expansion

$$\begin{aligned} [T, Y(A, z)]f(x) &= \frac{d}{dx} (A(x+z)f(x)) - A(x+z) \frac{d}{dx} f(x) \\ &= \left(\frac{d}{dx} A(x+z) \right) f(x) = \left(\frac{\partial}{\partial z} Y(A, z) \right) f(x) \end{aligned}$$

This then lets us justify calling T a *translation operator*.

- Since the vertex operators are just shift operators,

$$A(x+z)B(x+z) = B(x+z)A(x+z)$$

all of the operators are local with $N \equiv 0$.

Therefore, real analytic functions on $\mathbb{R}$ with $T = \frac{d}{dx}$ form a vertex algebra. Likewise, any vertex algebra such that for all A and B

$$Y(A, z)Y(B, w) = Y(B, w)Y(A, z)$$

is just a commutative algebra.

6.2. The Heisenberg Algebra. A very important example of a vertex algebra can be derived from the *Heisenberg algebra*, a Lie algebra with generators $\{b_n\}_{n \in \mathbb{Z}}$ and relations

$$[b_n, b_m] = n\delta_{n, -m} \mathbf{1} \quad [\mathbf{1}, b_n] = 0.$$

This is basically the definition of an affine Lie algebra over $\mathfrak{u}(1)$ (but note that $\mathfrak{u}(1)$ is not semisimple). For $n \geq 0$ we can identify b_n with an “annihilation operator”, and for $n < 0$ we can identify b_n with a “creation operator”.

We can associate to the Heisenberg Lie algebra a *vacuum representation*, where b_n acts as zero on the vacuum. Since we can compose operators in this representation, this gives us for free a representation of the *universal enveloping algebra*, the vector space of formal products of elements satisfying

$$m_1 m_2 - m_2 m_1 = [m_1, m_2]$$

for any two elements in the Heisenberg Lie algebra. It is this representation π (known as the *Fock space*) that will be the vector space of our vertex algebra, and the vacuum state $|0\rangle$ is just the $\mathbf{1}$ of the Heisenberg algebra. By the Poincaré–Birkhoff–Witt theorem, the universal enveloping algebra has a vector space basis of monomials $\prod_i b_i$ with increasing i . This gives us a basis for π . We can then assign to each monomial a *degree* given by

$$\deg \left(\prod_i b_i \right) = - \sum_i i \quad \deg(\mathbf{1}) = 0$$

Therefore, π is $\mathbb{Z}$ -graded (actually $\mathbb{N}$ -graded). We can then define the translation operator T inductively by degree. We first define

$$T\mathbf{1} = 0 \quad [T, b_n] = -nb_{n-1}$$

and then for any monomial m

$$T(b_k m) = b_k Tm + [T, b_k]m$$

and T extends by linearity to all of π .

Next, we can define our state-to-field operators. As it turns out, these are completely determined by the assignment

$$Y(b_{-1}, z) = \sum_{n \in \mathbb{Z}} b_n z^{-n-1}$$

and by locality. It can be verified that these assignments actually satisfy the vertex algebra axioms, but this will not be done here (see [4]).

6.3. The Virasoro Algebra. Vertex algebras originally arose in the study of *conformal field theory*, quantum field theories that are invariant under *conformal transformations*.

Definition 6.4. A *conformal transformation* is a transformation that locally preserves angles.

The Lie group of conformal transformations in two dimensions is generated by the Lie algebra of vector fields of the form

$$L_n = -z^{n+1}\partial_z$$

with $z \in \mathbb{C} \simeq \mathbb{R}^2$ [2]. For example, L_{-1} generates translations in the plane, which are conformal. These generators form the *Witt algebra*.

$$[L_m, L_n] = (m - n)L_{m+n}$$

A conformal field theory must therefore have an action of a projective representation of the Witt algebra. This means they must admit a representation of the unique central extension of the Witt algebra, the *Virasoro algebra*.

$$\begin{aligned} [L_m, L_n] &= (m - n)L_{m+n} + \frac{\hat{c}}{12}(m^3 - m)\delta_{m+n,0} \\ [\hat{c}, L_n] &= [\hat{c}, \hat{c}] = 0 \end{aligned}$$

where $\hat{c}$ is a new generator known as the *central charge*. The factor of $\frac{1}{12}$ may seem arbitrary (because it is), but it's related to derivations of the *Casimir effect* in conformal field theory. In this sense (comparing it to more traditional derivations), it's related to the famous

$$\zeta(-1) = "1 + 2 + 3 + \dots" = -\frac{1}{12}$$

Vertex algebras end up being a natural way to encode the structure of the Virasoro algebra.

Definition 6.5. A *vertex operator algebra* (VOA) is a $\mathbb{Z}$ -graded vertex algebra equipped with a *conformal element* ω such that

- $\deg(\omega) = 2$
- Writing

$$Y(\omega, z) = \sum_{n \in \mathbb{Z}} L_n^V z^{-n-1}$$

we have that $L_{-1}^V = T$ and $L_2^V \omega = \frac{c}{2} |0\rangle$, where c is the *central charge*.

We then automatically get that

Miracle 6.6. The L_n^V defined above satisfy the relations of the Virasoro algebra.

Example 6.7. The Heisenberg algebra has a one-parameter family of conformal elements

$$\omega_\lambda = \frac{1}{2}b_{-1}^2 + \lambda b_{-2}$$

with associated central charges $c_\lambda = 1 - 12\lambda^2$.

6.4. VOAs and Affine Lie Algebra. As it turns out,

Theorem 6.8. *Every affine Lie algebra can be given a vertex algebra structure.*

This proceeds in a similar manner to the Heisenberg case. We first define a vacuum representation of the affine Lie algebra $\mathfrak{g}$. We identify coefficients of the loop algebra of t^n for $n \geq 0$ as annihilation operators and coefficients of t^n for $n < 0$ as creation operators. Our vacuum representation is then a representation of the universal enveloping algebra of $\mathfrak{g}$ such that coefficients of t^n act trivially and K acts a scalar.

We then have that vacuum state $|0\rangle = \mathbf{1}$ is the identity in the universal enveloping algebra. We define an identical $\mathbb{Z}$ -grading and translation operator

$$T\mathbf{1} = 0 \quad [T, J_n^\alpha] = -nJ_{n-1}^\alpha$$

and vertex operators fixed by

$$Y(J_{-1}^a \mathbf{1}, z) = \sum_{n \in \mathbb{Z}} J_n^a z^{-n-1}$$

This defines a vertex algebra out of any affine Lie algebra.

7. LATTICE VOAs

7.1. Constructing Lattice VOAs. We've already seen a connection between lattices and affine Lie algebras and a connection between affine Lie algebras and vertex algebra. However, there's a more direct way to turn lattices into vertex algebras.

For an n -dimensional lattice, we start by constructing an n -dimensional version of the Heisenberg algebra. Just as the Heisenberg algebra acts like the affinization of $\mathfrak{u}(1)$, we will construct an affinization of $\mathfrak{u}(1)^n$. This is a Lie algebra with generators h_k^i for $i \in \{1, 2, \dots, n\}$, $k \in \mathbb{Z}$ and brackets

$$(7.1) \quad [h_n^a, h_m^b] = n\delta^{ab}\delta_{n,-m}K$$

where K is central. Like before, we can let S be the subalgebra of the universal enveloping algebra generated by elements of the form h_n^i where $n < 0$.

We're not done. Given any even lattice, we can construct the *group algebra* $\mathbb{C}[\Lambda]$, the associative algebra with generators $\{|\lambda\rangle\}_{\lambda \in \Lambda}$ and multiplication

$$|\lambda_1\rangle \cdot |\lambda_2\rangle = |\lambda_1 + \lambda_2\rangle$$

Finally, we take our vector space to be $S \otimes \mathbb{C}[Q]$, that is, the vector space generated by elements of the form $s|\lambda\rangle$, with $s \in S$ and $\lambda \in \Lambda$. We once again have a vacuum vector $|0\rangle$ and an inductively-defined translation operator T .

Exactly like with the Heisenberg VOA, we can construct fields

$$Y(h_{-1}^i |0\rangle, z) = \sum_{n \in \mathbb{Z}} h_n^i z^{-n-1}$$

corresponding to each of our generators h_{-1}^i . The construction of the fields $Y(|\lambda\rangle, z)$ is considerably more complicated (see [6]), but once defined generate a vertex algebra from any even lattice. In string theory, for a given Λ , this represents a theory of bosons on the torus $\mathbb{R}^n/\Lambda$. (If the lattice is not even, the resulting structure is instead a vertex *superalgebra*, which is a modified form of a vertex algebra.)

We have a $\mathbb{Z}$ -grading given by

$$\deg(m|\lambda\rangle) = \deg(m) + \frac{1}{2} \langle \lambda, \lambda \rangle$$

for m in the Heisenberg algebra. We can turn this into a VOA with the conformal vector

$$\omega = \frac{1}{2} \sum_{i=1}^r h_{-1}^i h_{-1}^i |0\rangle$$

Theorem 7.2. *The lattice VOA defined above has central charge $c = \text{rank}(\Lambda)$.*

7.2. The Leech Lattice. Of all possible lattices in all possible Euclidean spaces, there is one that is particularly special.

Definition 7.3. The *Leech lattice* is the unique even unimodular lattice in $\mathbb{R}^{24}$ such that all nonzero vectors have norm at least 2.

The Leech lattice has many exceptional properties. For example, it gives the optimal packing of 24-dimensional spheres. It can be constructed in many ways, but the easiest is just to write out the generators explicitly as in [7]. What's immediately relevant is that, since the Leech lattice is even, we can construct its corresponding VOA V_{Leech} . This has central charge

$$c = \text{rank}(\Lambda_{\text{Leech}}) = 24$$

7.3. Characters of Lattice VOAs. We can define the character of a VOA V with central charge c in a way that generalizes the definition for affine Lie algebras. Recall that every VOA has a grading $V = \bigoplus_n V_n$ given by the eigenvalues of L_0 . Then we can define

$$\chi_V = \text{tr}_V \left(q^{L_0 - c/24} \right) = q^{-c/24} \sum_n \dim V_n q^n$$

Example 7.4. Consider the Heisenberg VOA with $\lambda = 0$ (so $c = 1$). Then $\dim V_n$ is simply the number of basis vectors with degree n . For example, when $n = 3$ there are 3 basis vectors.

$$b_{-1}^3 \quad b_{-2}b_{-1} \quad b_{-3}$$

In general, the number of basis vectors is given by the number of *partitions* of n . We can write the corresponding generating function as

$$\sum_n P(n) q^n = \prod_n (1 + q^n + q^{2n} + \cdots) = \prod_n \frac{1}{1 - q^n}$$

so

$$\chi_V = \frac{1}{q^{1/24} \prod_n (1 - q^n)} = \frac{1}{\eta(q)}$$

Theorem 7.5. *The character formula for lattice VOAs gives*

$$\chi_\Lambda = \frac{\Theta_\Lambda}{\eta^{\text{rank}(\Lambda)}}$$

Proof. We can simply compute

$$\begin{aligned}
\chi_\Lambda(q) &= q^{-\text{rank}(\Lambda)/24} \sum_{s \in S, \lambda \in \Lambda} q^{\deg(S) + \frac{1}{2} \langle \lambda, \lambda \rangle} \\
&= \Theta_\Lambda(q) \cdot \frac{1}{q^{\text{rank}(\Lambda)/24}} \left(\sum_n P(n) q^n \right)^{\text{rank}(\Lambda)} \\
&= \frac{\Theta_\Lambda(q)}{\eta(q)^{\text{rank}(\Lambda)}}
\end{aligned}$$

□

Remark 7.6. We notice that if $\Lambda = \sqrt{2}\mathbb{Z}$ is (up to a scaling convention) the root lattice of $A_1^{(1)}$, this formula is precisely the formula for the character of the level 1 representation of $A_1^{(1)}$. This is not a coincidence. It was an early result of this field that if the lattice is the root lattice of an affine Lie algebra,⁵ then the resulting lattice VOA is a concrete construction of a level 1 representation of that Lie algebra. In particular the generators h_k^i correspond to the generators of the Cartan subalgebra, while for a root λ , E^λ corresponds to the operator

$$e^{i\lambda \cdot q}.$$

Finally L_0 is essentially the sum

$$L_0 = \frac{1}{2} \sum_{i=1}^d \sum_{n \in \mathbb{Z}} h_{-n}^i h_n^i$$

(as written this sum does not converge when evaluated at $|0\rangle$, so technically one needs to “normal order” this sum, however we will ignore this). Then using the (7.1) and standard identities we can compute

$$\begin{aligned}
[L_0, h_k^j] &= \frac{1}{2} \sum_{i=1}^d \sum_{n \in \mathbb{Z}} [h_{-n}^i h_n^i, h_k^j] = \frac{1}{2} \sum_{i=1}^d \sum_{n \in \mathbb{Z}} h_{-n}^i [h_n^i, h_k^j] + [h_{-n}^i, h_k^j] h_n^i \\
&= \frac{1}{2} (h_n^j n + n h_n^j) = n h_n^j
\end{aligned}$$

so indeed L_0 has the same relations as the operator D , so we expect the trace of L_0 (which is the character in the VOA sense) to correspond to the trace of D (which is the basic specialization of the character in the affine Lie algebra sense). Thus with the power of VOAs we derive the much more powerful version of [Theorem 5.21](#).

Corollary 7.7. *Let $\mathfrak{g}$ be a (simply-laced) affine Lie algebra simple root lattice Λ of rank r (under the right scaling convention). Then the basic specialization of the character of the level 1 representation is*

$$\tilde{\chi}(0, \tau, 0) = \frac{\Theta_\Lambda(\tau)}{\eta(\tau)^r}.$$

Back to the Leech lattice, we observe that the character of V_{Leech} is simply

$$\frac{\Theta_{\text{Leech}}}{\eta^{24}}$$

⁵In particular, a *simply-laced* affine Lie algebra, which is to say that all of the roots have the same norm.

Note that this is a weight 0 meromorphic modular form, which means it must be a rational function of the j -invariant! Observing also that there are no singularities, we can simply compute out the first two terms

$$\frac{\Theta_{\text{Leech}}}{\eta^{24}} = q^{-1} + 24 + \dots$$

and deduce that the rational function must be

$$\chi_{\text{Leech}} = j - 720$$

Hence, we see a direct connection between the Leech lattice and the j -invariant!

8. MOONSHINE FOR FINITE SIMPLE GROUPS

8.1. The Classification of Finite Simple Groups. A *simple group* is a group with no nontrivial proper normal subgroups.

Question 8.1. What are all of the finite simple groups?

Theorem 8.2. (*the Classification of Finite Simple Groups*) Every finite simple group is one of the following:

- A cyclic group C_p of prime order
- An alternating group A_n , $n \geq 5$
- One of 16 infinite families known as the *groups of Lie type*
- One of 26 *sporadic* groups which don't fit into any infinite family

Sporadic simple groups often arise as automorphisms of exceptional structures. For example,

- Take the set $X = \{1, 2, \dots, 24\}$ and consider *octads*, 8-element subsets of X . Then there exists a set of 759 octads such that each 5-element subset of X occurs in exactly one octad. This is called the *Steiner system* $S(5, 8, 24)$. The automorphism group of $S(5, 8, 24)$ is the Mathieu group M_{24} , a sporadic simple group.
- The automorphism group of the Leech lattice Λ is not simple since the automorphism -1 generates the center of $\text{Aut}(\Lambda)$. However, the quotient of $\text{Aut}(\Lambda)$ by its center is the Conway group Co_1 , a sporadic simple group.

The largest sporadic simple group is the *Monster group* M , with order⁶

808,017,424,794,512,875,886,459,904,961,710,757,005,754,368,000,000,000.

Does it arise similarly?

8.2. The Moonshine Module.

Theorem 8.3. The *Monster group* is the automorphism group of the $V^\natural$ (“ V natural”) vertex algebra.

We can construct $V^\natural$ starting from the Leech lattice VOA V_Λ by the *orbifold construction*.

Definition 8.4. An *orbifold* is a space that is locally homeomorphic to the quotient of $\mathbb{R}^n$ by a discrete group action Γ

⁶It should be noted that this is smaller than the order of the permutation group of a 52-card deck of cards.

The orbifold construction allows us to turn a string theory on a manifold S into a string theory on S/Γ . In our case, observe that the -1 automorphism of the Leech lattice induces an automorphism of V_Λ . We denote the subalgebra fixed by this automorphism $V_\Lambda(0)$. We can also construct a “twisted” version of this algebra $V_{\Lambda,-1}(0)$ and finally construct

$$V^\natural = V_\Lambda(0) \oplus V_{\Lambda,-1}(0)$$

which describes a string theory on $(\mathbb{R}^{24}/\Lambda)/\{\pm 1\}$ with central charge $c = 24$. After detailed computation, we have both that

$$\text{Aut}(V^\natural) = M \text{ and } \chi_{V^\natural} = j(\tau) - 744$$

This solves the original mystery. Recall that the coefficients of the character of a vertex algebra V give the dimensions of subspaces of V . For example, the q^1 coefficient of $\chi_{V^\natural}$ tells us there is a 196884-dimensional space $V_2^\natural$ spanned by vectors in $V^\natural$ with degree 2. The action of M on $V^\natural$ turns this into a nontrivial 196884-dimensional representation of M , which must decompose into the direct sum of the smallest nontrivial M -representation, which has dimension 196883, and a trivial 1-dimensional M -representation. In other words,

$$196884 = 196883 + 1$$

It is conjectured that $V^\natural$ is the *unique* vertex algebra with $c = 24$ and no elements of degree 1, which would help explain why the Monster is so special and why $V^\natural$ is so natural.

8.3. Generalizations. Although this specific relation between M and j has become famous, the full monstrous moonshine conjecture (originally by Conway and Norton and proven by Borchers) was stronger. We’ve been working mostly with characters *at the identity*, which are just the dimensions of spaces. If we instead replace the dimension with other characters of $V^\natural$, we get a *McKay-Thompson series* of $V^\natural$.

$$T_g(q) = \sum_{n=-1}^{\infty} \chi_{V^\natural_n}(g) q^n$$

It turns out that these are all very special functions called *Hauptmoduln*. Conway and Norton conjectured the existed of a representation like $V^\natural$ based on this observation.

Additionally, there is ongoing work to derive Moonshine-like results for other sporadic simple groups, the first new example being *Mathieu moonshine*. We have that in addition to the Leech lattice, there are 23 other even unimodular lattices in $\mathbb{R}^{24}$ called the *Niemeier lattices*. Just as monstrous moonshine is a connection between M , the Leech lattice and modular forms, Mathieu moonshine is a connection between M_{24} , the Niemeier lattice A_1^{24} , and *mock modular forms*, which are generalizations of modular forms. Other examples of moonshine, known collectively as *umbral moonshine*, have been worked out for other sporadic simple groups.

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